

# **THE GÖDEL INCOMPLETENESS THEOREMS**

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Everything in the lectures or on the problem sheets is on the syllabus and examinable, unless otherwise indicated.\*

Prerequisites: an introductory course in logic is assumed.

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\* Anything in the footnotes is not on the syllabus.

## 0. Introduction

We will usually assume that the semiring  $(\mathbb{N}, 0, 1, +, \cdot)$  exists and has the usual familiar properties (from which it will follow that various axiom systems for it are consistent, since they have a model).

I assume familiarity with the Completeness Theorem of first-order logic; so when I prove a statement such as  $S \vdash \phi$  ( $\phi$  is formally provable from assumptions  $S$ ) I will on the whole not provide a formal proof of  $\phi$  from  $S$ ; I will instead argue that such a formal proof exists (which is quite different and much easier). I also assume some skill in distinguishing language from metalanguage and theorems from metatheorems.

These lectures are based on lecture notes by Dan Isaacson, and on Raymond Smullyan's book *Gödel's Incompleteness Theorems* (OUP, 1992). However, I sometimes depart (in notation or in other respects) from both sources.

## 1. A formal language for arithmetic

### 1.1. The language itself

We choose a formal language to make Gödel numbering more straightforward.

DEFINITION 1.1.1. allies *The symbols of the language  $\mathcal{L}_E$  are:*

$$\bar{0} \ + \ v \ f' \ ( \ ) \ \neg \ \rightarrow \ \forall \ = \ \leq \ \#$$

*An expression in  $\mathcal{L}_E$  is any finite, non-empty sequence of symbols of  $\mathcal{L}$  that does not begin with  $+$ .*

The rules of syntax are as follows.

DEFINITION 1.1.2. The *The terms of  $\mathcal{L}_E$  are defined as follows.*

$\bar{0}$  is a numeral term, and if  $\sigma$  is a numeral term, then so is  $\sigma^+$ . We will write  $\bar{0}$  followed by  $n^+$ 's as  $\bar{n}$ .

$v$  is a variable term, and if  $\tau$  is a variable term, then so is  $\tau'$ . If  $n$  is a natural number (including zero), then we'll write  $v_n$  for  $v$  followed by  $n^+$ 's.

The function labels are  $f$ ,  $f'$ , and  $f''$ .

A term is a numeral term, a variable term, an expression  $\sigma^+$  where  $\sigma$  is a term, or an expression  $(\tau \sigma v)$ , where  $\sigma$  is a function label and  $\tau$  and  $v$  are terms.

DEFINITION 1.1.3. edge  $\mathcal{L}_E$  contains the following formulae.

An atomic formula is an expression  $\sigma = \tau$  or  $\sigma \leq \tau$ , where  $\sigma$  and  $\tau$  are terms.

Other formulae are:  $\neg\phi$ ,  $(\phi \rightarrow \psi)$ ,  $\forall x \phi$ , where  $\phi$  and  $\psi$  are formulae, and  $x$  is a variable term.

We sometimes write  $(\phi \vee \psi)$  for  $(\neg\phi \rightarrow \psi)$ ,  $(\phi \wedge \psi)$  for  $\neg(\neg\phi \vee \neg\psi)$ ,  $(\phi \leftrightarrow \psi)$  for  $((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))$ , and  $\exists v_i \phi$  for  $\neg\forall v_i \neg\phi$ , where  $\phi$  and  $\psi$  are formulae, and  $v_i$  is a variable term.

DEFINITION 1.1.4. of If  $x$  and  $y$  are finite sequences of symbols in  $\mathcal{L}_E$ , then their concatenation  $x \frown y$  is the sequence obtained by adding  $y$  after  $x$ .

DEFINITION 1.1.5. war  $\mathcal{L}$  is the sublanguage of  $\mathcal{L}_E$  containing no occurrences of  $f''$ .

## 1.2. Logical rules

Given the Completeness Theorem of first-order Predicate Calculus, it does not much matter which system of axioms and logical rules we use. We choose to use one which makes it easier to prove the (meta)theorems we want to use (but which is also difficult to use for constructing formal proofs).

So, we use the following axiom schemes:

DEFINITION 1.2.1. like The logical axioms are all instances of the following schemata, where  $\phi$ ,  $\chi$  and  $\psi$  may be any formulae:

(A1)  $(\phi \rightarrow (\chi \rightarrow \phi))$

(A2)  $((\phi \rightarrow (\chi \rightarrow \psi)) \rightarrow ((\phi \rightarrow \chi) \rightarrow (\phi \rightarrow \psi)))$

(A3)  $((\neg\phi \rightarrow \neg\chi) \rightarrow (\chi \rightarrow \phi))$

(A4)  $(\forall v_i \phi(v_i) \rightarrow \phi(t))$ , where  $v_i$  is a variable letter and  $t$  is a term which can be sensibly substituted for  $v_i$ , that is, it contains no variable letter  $v_j$  such that  $v_i$  occurs free in  $\phi$  in the scope of a quantifier  $\forall v_j$ ,

(A5)  $(\forall v_i (\phi \rightarrow \chi) \rightarrow (\phi \rightarrow \forall v_i \chi))$ , for  $v_i$  a variable letter, provided  $v_i$  does not occur free in  $\phi$ ,

(A6)  $\forall v_i (v_i = v_i)$ ,

(A7) if  $F$  and  $G$  are atomic formulae, where  $G$  is obtained from  $F$  by replacing some, but not necessarily all, occurrences of  $v_i$  by  $v_j$ , then  $((v_i = v_j) \rightarrow (F \rightarrow G))$ .

DEFINITION 1.2.2. an The rules of inference are the following, where  $\phi$  and  $\chi$  are any formulae and  $x$  is any variable letter:

(MP) Modus Ponens: that is, from  $\phi$  and  $\phi \rightarrow \chi$  deduce  $\chi$ ;

(Gen) Generalisation: from  $\phi$  deduce  $\forall v_i \phi$ .

This system is slightly different from that in the B1.1 Logic course. However the Completeness Theorem is true for both, and they have exactly the same sets of theorems.

DEFINITION 1.2.3. ill If  $\Gamma$  is a (possibly empty) set of formulae, and  $\phi$  is a formula, we say that  $\phi$  can be proved from  $\Gamma$ , and write  $\Gamma \vdash \phi$ , if and only if there exists a finite sequence  $\phi_1, \dots, \phi_n$  of formulae such that  $\phi_n = \phi$ , and for each  $i$ ,  $\phi_i$  is an element of  $\Gamma$ , or a logical axiom, or else it is deduced from previous members of the sequence using a rule of inference.

We will need to refer to the details of the system occasionally.

## 1.3. Interpretation

We will usually interpret  $\mathcal{L}$  as applying to the semiring  $(\mathbb{N}, 0, 1, +, \cdot)$ , where  $\bar{0}$  is interpreted as referring to 0,  $+$  as referring to the successor function  $n \mapsto n + 1$  (so that  $\bar{n}$  refers to  $n$ ), and the function symbols  $f$ ,  $f'$  as referring, respectively, to addition and multiplication; and we interpret  $\mathcal{L}_E$  as referring to the expansion obtained by adding the exponentiation operation, when  $f''$  will refer to exponentiation.

For terms  $\sigma$  and  $\tau$ , we normally rewrite  $(\sigma f \tau)$  as  $\sigma + \tau$ ,  $(\sigma f' \tau)$  as  $\sigma \cdot \tau$ , and  $(\sigma f'' \tau)$  as  $\sigma^\tau$ .

We will normally define truth with respect to this interpretation, though we will sometimes remember to say “true in  $\mathbb{N}$ ” to make this a little clearer. We will occasionally refer to other interpretations.

DEFINITION 1.3.1. sheathed A subset  $A$  of  $\mathbb{N}^k$  is definable if and only if there exists a formula  $\phi(v_1, \dots, v_k)$  with only  $v_1, \dots, v_k$  free, such that  $\phi(\overline{n_1}, \dots, \overline{n_k})$  is true if and only if  $(n_1, \dots, n_k) \in A$ . We say that  $A$  is provably definable from a set of assumptions  $S$  if  $S \vdash \phi(\overline{n_1}, \dots, \overline{n_k})$  if  $(n_1, \dots, n_k) \in A$ , and  $S \vdash \neg\phi(\overline{n_1}, \dots, \overline{n_k})$  if  $(n_1, \dots, n_k) \notin A$ .

A function  $g : \mathbb{N}^k \rightarrow \mathbb{N}$  is definable if and only if the set  $A = \{(n_1, \dots, n_k, g(n_1, \dots, n_k)) : n_1, \dots, n_k \in \mathbb{N}\}$  is definable, and is weakly provably definable from a set of assumptions  $S$  if  $A$  is provably definable.  $f$  is provably definable from  $S$  if for all  $n_1, \dots, n_k \in \mathbb{N}$ ,  $S \vdash \forall v_1 (\phi(\overline{n_1}, \dots, \overline{n_k}, v_1) \leftrightarrow v_1 = g(n_1, \dots, n_k))$ , where  $\phi$  is the formula defining  $A$ .

## 2. Peano arithmetic

### 2.1. The Peano axioms

We will be considering a number of axiom schemes for arithmetic on  $\mathbb{N}$  of different strengths. The most famous, and most commonly used, is:

DEFINITION 2.1.1. knife We will denote by PA (Peano Arithmetic) the following list of statements (all of which are expressible in  $\mathcal{L}$ ):

1.  $\forall v_i \neg v_i^+ = \overline{0}; \forall v_i \forall v_j (v_i^+ = v_j^+ \rightarrow v_i = v_j)$ .  
( $n \mapsto n^+$  is an injection from  $\mathbb{N} \leftrightarrow \mathbb{N} \setminus \{0\}$ ).
2.  $\forall v_i v_i + \overline{0} = v_i$  and  $\forall v_i v_i \cdot \overline{0} = \overline{0}$ .
3.  $\forall v_i \forall v_j v_i + v_j^+ = (v_i + v_j)^+$  and  $\forall v_i \forall v_j v_i \cdot v_j^+ = (v_i \cdot v_j) + v_i$ .
4. i.  $\forall v_i \overline{0} \leq v_i; \forall v_i \forall v_j (v_i \leq v_j \leftrightarrow (v_i = v_j \vee v_i^+ \leq v_j))$ ;  
ii.  $\forall v_i v_i \leq v_i$ ;  
iii.  $\forall v_i \forall v_j ((v_i \leq v_j) \wedge (v_j \leq v_i)) \rightarrow v_i = v_j$ ;  
iv.  $\forall v_i \forall v_j \forall v_k ((v_i \leq v_j \wedge v_j \leq v_k) \rightarrow (v_i \leq v_k))$ ;  
v.  $\forall v_i \forall v_j (v_i \leq v_j \vee v_j \leq v_i)$ .

( $\leq$  is a total order, with initial element  $\overline{0}$ , and  $n^+$  is the immediate successor of  $n$ .)

5. (Induction Schema): For any formula  $\phi(v_1)$  of  $\mathcal{L}$ , the following is an axiom: if  $\phi(\overline{0})$ , and if for all  $n$ ,  $\phi(n)$  implies  $\phi(n^+)$ , then  $\forall n \phi(n)$ .

Formally:

$$((\phi(\overline{0}) \wedge (\forall v_1 \phi(v_1) \rightarrow \phi(v_1^+))) \rightarrow \forall v_1 \phi(v_1)).$$

EXERCISE 2.1.2. no From PA it is provable that the successor function  $n \mapsto n^+$  is onto  $\mathbb{N} \setminus \{0\}$ .

EXERCISE 2.1.3. more From PA it is provable that if  $m$  and  $n$  are natural numbers, then  $m \leq n$  iff  $\exists k m + k = n$ .

The strongest axiom set for arithmetic we'll be using is the following.

DEFINITION 2.1.4. shall We will denote by PAE (Peano Arithmetic with Exponentiation) PA, augmented by the following statements:

- 2':  $\forall v_i v_i^{\overline{0}} = \overline{1}$ .
- 3':  $\forall v_i, v_j v_i^{v_j^+} = v_i^{v_j} \cdot v_i$ .

5': instances of the induction schema involving formulae  $\phi(v_1)$  belonging to  $\mathcal{L}_E$  but not to  $\mathcal{L}$ .

## 2.2. Gödel numbering

NOTATION 2.2.1. cut We will from time to time write numbers in base 13. When we do that, we will use the symbol  $A$  to refer to ten,  $B$  to refer to eleven, and  $C$  to refer to twelve.

When confusion is likely, we'll use a subscript  $_{13}$  to indicate that a number is to be read in base 13, and  $_{10}$  to indicate that it should be read in base 10.

The alphabet of  $\mathcal{L}$  has thirteen symbols, and we will assign numbers 0 to 12 to them. A string has a Gödel number, which is got by replacing each symbol by a digit in the set  $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C$ , and then interpreting the result as a number in base 13.

(Thirteen is convenient partly because it's prime, and partly because with more symbols, it's easier to work out how to write stuff. We could get away with two symbols, by representing each of the above thirteen symbols by a different string of four 0's and 1's.)

More formally:

DEFINITION 2.2.2. his Gödel numbers are assigned to the symbols of the language  $\mathcal{L}_E$  as follows:

|     |     |     |     |     |     |     |        |               |           |     |        |      |
|-----|-----|-----|-----|-----|-----|-----|--------|---------------|-----------|-----|--------|------|
| $+$ | $0$ | $($ | $)$ | $f$ | $'$ | $v$ | $\neg$ | $\rightarrow$ | $\forall$ | $=$ | $\leq$ | $\#$ |
| 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7      | 8             | 9         | A   | B      | C    |

—where all numbers are to be read in base 13. If  $s$  is a symbol of  $\mathcal{L}_E$ , then its Gödel number may be written as  $\ulcorner s \urcorner$ .

The Gödel number of an expression of  $\mathcal{L}_E$  is obtained by writing the Gödel numbers of the individual symbols in order, and reading the result in base 13; that is, if  $\phi = s_0 s_1 \dots s_r$ , where the  $s_i$  are symbols of  $\mathcal{L}$  and  $s_0$  is not  $+$ , then:

$$\ulcorner \phi \urcorner = (\ulcorner s_0 \urcorner \ulcorner s_1 \urcorner \dots \ulcorner s_r \urcorner)_{13}.$$

One can quickly convince oneself that

$$(\ulcorner s_0 \urcorner \ulcorner s_1 \urcorner \dots \ulcorner s_r \urcorner)_{13} = \ulcorner s_0 \urcorner 13^r + \ulcorner s_1 \urcorner 13^{r-1} + \dots + \ulcorner s_r \urcorner.$$

DEFINITION 2.2.3. master The Gödel number of a term or formula is defined as in the previous definition. The Gödel number of a sequence of terms or formulae is obtained by separating the formulae by  $\#$ , so that

$$\ulcorner (\phi_1, \dots, \phi_k) \urcorner = \ulcorner \# \phi_1 \# \phi_2 \# \dots \# \phi_k \# \urcorner.$$

NOTE: All we really require of our system of Gödel numbering is that there should exist a definable (in  $\mathcal{L}_E$ ) function  $\circ$  such that  $\ulcorner \phi \psi \urcorner = \ulcorner \phi \urcorner \circ \ulcorner \psi \urcorner$ , and that the function  $n \mapsto \ulcorner \bar{n} \urcorner$  should be definable.

We commit the abuse of using symbols such as  $x, y, m, n$  and so forth for  $v_0, v_1$  and so forth.

### 2.3. The arithmetical hierarchy

We classify formulae in prenex normal form according to the string of quantifiers at the front.

**DEFINITION 2.3.1.** therefore A bounded quantifier is of the form  $\exists m \leq n$  or  $\forall m \leq n$  (strictly not in our language, but  $\exists m \leq n \phi$  can be expressed by  $\exists m (m \leq n \wedge \phi)$ , and  $\forall m \leq n \phi$  can be expressed by  $\forall m (m \leq n \rightarrow \phi)$ .)

**DEFINITION 2.3.2.** friends A formula is  $\Sigma_0$ ,  $\Pi_0$ , or  $\Delta_0$ , if it contains no unbounded quantifiers.

If  $\phi$  is  $\Sigma_n$ , then  $\forall m \phi$  is  $\Pi_{n+1}$ , and if  $\phi$  is  $\Pi_n$ , then  $\exists m \phi$  is  $\Sigma_{n+1}$ .

We say that a formula  $\phi$  is provably  $\Sigma_n$  (or)  $\Pi_n$  if there is a formula  $\phi'$  which is respectively  $\Sigma_n$  or  $\Pi_n$ , such that  $\phi \leftrightarrow \phi'$  is a theorem. If  $S$  is a set of axioms, then we say  $\phi$  is provably  $\Sigma_n$  or  $\Pi_n$  with respect to  $S$  if there is a formula  $\phi'$  which is respectively  $\Sigma_n$  or  $\Pi_n$ , such that  $S \vdash \phi \leftrightarrow \phi'$ . If  $\phi$  is provably  $\Sigma_n$  and provably  $\Pi_n$ , then we say that it is  $\Delta_n$ ; similarly with  $\Delta_n$  with respect to  $S$ .

We often omit the word “provably”.

**EXAMPLE 2.3.3.** as As an example,  $(\neg m \leq n \vee \exists k m + k = n)$  is not  $\Sigma_0$ , but is provably  $\Sigma_0$ , since it is provably equivalent to  $\exists k \leq n (\neg m \leq n \vee m + k = n)$ .

**PROPOSITION 2.3.4.** far The set of formulae that is provably  $\Sigma_n$  is closed under conjunction, disjunction, bounded quantification and, if  $n > 0$ , existential quantification.

The set of formulae that is provably  $\Pi_n$  is closed under conjunction, disjunction, bounded quantification and, if  $n > 0$ , universal quantification.

**PROOF:** Exercise.  $\square$

### 2.4. Results concerning expressibility

**DEFINITION 2.4.1.** as A set  $T$  of expressions of  $\mathcal{L}_E$  is definable if and only if there exists a formula  $\phi(x)$  such that  $\phi(\ulcorner \psi \urcorner)$  is true if and only if  $\psi$  belongs to  $T$ .

If  $S$  is a set of sentences of  $\mathcal{L}_E$ , and  $T$  is a set of expressions, we will say that  $T$  is provably definable from  $S$  if for some formula  $\phi$ ,  $\phi(\ulcorner \psi \urcorner)$  is provable from  $S$  if and only if  $\psi$  belongs to  $T$ , and  $\neg \phi(\ulcorner \psi \urcorner)$  is provable from  $S$  if and only if  $\psi$  does not belong to  $T$ .

**DEFINITION 2.4.2.** to A property  $\phi$  of natural numbers or of finite sequences of natural numbers is expressible iff the set  $\{n : \mathbb{N} \models \phi(\bar{n})\}$  is definable.

**PROPOSITION 2.4.3.** the Any finite set of expressions is definable, and is indeed provably definable (from  $\emptyset$ ).

**LEMMA 2.4.4.** sepulchre Express  $m < n$  as  $m \leq n \wedge \neg m = n$ . This is  $\Sigma_0$ .  
 $m \mid n$  is provably  $\Sigma_0$ .

Write  $[m, n]$  for  $\frac{1}{2}(m+n+1)(m+n)+m$ . The function  $(m, n) \mapsto [m, n]$  is a bijection between  $\mathbb{N} \times \mathbb{N}$  and  $\mathbb{N}$ , and the statement “ $k = [m, n]$ ” is provably  $\Sigma_0$ .

**LEMMA 2.4.5.** of The statement that  $r$  is the largest number such that  $13^r \leq n$  is expressible in  $\Sigma_0$ .

**PROOF:** We express it as follows:  $\overline{13}^r \leq \bar{n} \wedge \neg \overline{13}^{r+1} \leq \bar{n}$ .  $\square$

LEMMA 2.4.6. christ *The statement that  $n$  is the result of concatenating  $k$  and  $l$  is expressible in  $\Sigma_0$ .*

PROOF: We express the statement as follows:

$k \neq 0$ ; also  $l = 0$  and  $n = 13.k$ , or  $l \neq 0$  and there exists  $r \leq l$  such that  $r$  is the greatest number such that  $13^r \leq l$ , and  $n = 13^{r+1}k + l$ .  $\square$

It's straightforward to get concatenations of more than two.

LEMMA 2.4.7. whose *The following can be expressed in complexity  $\Sigma_0$ :*

1.  $k$  is an initial part of  $n$ ;
2.  $l$  is a final part of  $n$ ;
3.  $k$  is a substring of  $n$ .

PROOF:  $k$  is an initial part of  $n$ :  $0 < k$ , and either there exists  $l \leq n$  such that  $n = k.13^l$ , or for some  $l, m \leq n$ ,  $n$  is the result of concatenating  $k.13^l$  and  $m$ .

$l$  is a final part of  $n$ : there exists  $k \leq n$  such that  $n$  is the result of concatenating  $k$  and  $l$ . We write this as  $l E n$ .

$k$  is a substring of  $n$ : there exists  $l \leq n$  such that  $l$  is an initial part of  $n$  and  $k$  is a final part of  $l$ . We write this as  $k P n$ .

These are all provably  $\Sigma_0$ .  $\square$

LEMMA 2.4.8. soldier *The following can be expressed in complexity  $\Sigma_0$ :*

1. *The first element of the string with Gödel number  $n$  has Gödel number  $m$  (where  $n$  is not zero, and in this case necessarily,  $m$  is not zero);*
2. *The last element of the string with Gödel number  $n$  has Gödel number  $m$ .*

PROOF: The first element of the string with Gödel number  $n$  has Gödel number  $m$  can be expressed thus:  $0 < m, m < 13$ , and either there exists  $l \leq n$  such that  $n = m.13^l$ , for some  $k, l \leq n$ ,  $n$  is the result of concatenating  $m.13^l$  and  $k$ .

The last element of the string with Gödel number  $n$  has Gödel number  $m$ :  $m = 0$  and  $13 \mid n$ , or  $0 < m < 13$  and there exists  $k \leq n$  such that  $n$  is the result of concatenating  $k$  and  $m$ .

These statements are provably  $\Sigma_0$ .  $\square$

LEMMA 2.4.9. now *The following statement is expressible in complexity  $\Sigma_0$ :*

*$n$  codes a sequence of (non-empty) expressions, the last member of which is  $\sigma$ .*

PROOF: We may express “ $n$  codes a sequence of (non-empty) expressions, the last member of which is  $\sigma$ ” as follows:  $\sigma$  contains no  $\#$ , and either  $n$  results from concatenating  $\ulcorner \# \urcorner$ ,  $\ulcorner \sigma \urcorner$ , and  $\ulcorner \# \urcorner$ , or there exists  $a \leq n$  such that  $a$  has first and last characters  $\#$ , the string  $\#\#$  does not occur in  $a$ , and  $n$  results from concatenating  $\ulcorner a \urcorner$ ,  $\ulcorner \sigma \urcorner$  and  $\ulcorner \# \urcorner$ .

This is  $\Sigma_0$ .  $\square$

### 3. Recursive functions

In this section we pin down exactly what sets and functions can be described in complexity  $\Delta_1$  and  $\Sigma_1$ .



### 3.1. Recursive functions

DEFINITION 3.1.1. under The primitive recursive functions are the smallest class of functions from finite powers of  $\mathbb{N}$  to  $\mathbb{N}$  with the following properties.

1. The constant function  $n \mapsto 0$  is primitive recursive.
2. The successor function  $n \mapsto n + 1$  is primitive recursive.
3. For any positive integer  $k$ , for any  $i \leq k$ , the projection function  $(n_1, \dots, n_k) \mapsto n_i$  is primitive recursive.
4. Composition: the function  $h(n_1, \dots, n_k) = g(f_1(n_1, \dots, n_k), \dots, f_m(n_1, \dots, n_k))$  is primitive recursive, when  $g$  and all  $f_j$  are primitive recursive.
5. Primitive recursion:  $f$  is primitive recursive, where  $f(n_1, \dots, n_k, 0) = g(n_1, \dots, n_k)$ , and for all  $n$ ,  $f(n_1, \dots, n_k, n + 1) = h(n_1, \dots, n_k, n, f(n_1, \dots, n_k, n))$ , where  $g$  and  $h$  are primitive recursive.

EXAMPLE 3.1.2. whose The addition function  $A : (m, n) \mapsto m + n$  is primitive recursive.

PROOF: Let  $h(m, n, k) = k + 1$  (this is primitive recursive, since it is the composition of a projection function with the successor function).

Let  $g(m) = m$  (the identity on  $\mathbb{N}$  is a projection, so is primitive recursive).

Then for all  $m$ , we define  $A$  by primitive recursion so that  $A(m, 0) = g(m)$ , and for all  $m$  and  $n$ ,  $A(m, n + 1) = h(m, n, A(m, n))$ .  $\square$

EXAMPLE 3.1.3. blessed The modified subtraction function  $S$  defined so that  $S(m, n) = m - n$  if  $m \geq n$  and  $S(m, n) = 0$  if  $m < n$ , is primitive recursive.

Multiplication and exponentiation are primitive recursive.

We obtain the recursive partial functions by also using the minimalisation operator, which, given a function  $g$ , returns the least  $n$  such that  $g(n_1, \dots, n_k, n) = 0$  if there is one, and is undefinable otherwise.

DEFINITION 3.1.4. cross The recursive functions are the smallest class of partial functions from finite powers of  $\mathbb{N}$  to  $\mathbb{N}$  with the following properties, where we use the notation  $f \simeq g$  to mean “ $f$  and  $g$  have the same domain, and on that domain they are equal”.

1. The constant function  $n \mapsto 0$  is recursive.
2. The successor function  $n \mapsto n + 1$  is recursive.
3. For any positive integer  $k$ , for any  $i \leq k$ , the projection function  $(n_1, \dots, n_k) \mapsto n_i$  is recursive.
4. The function  $h(n_1, \dots, n_k) \simeq g(f_1(n_1, \dots, n_k), \dots, f_m(n_1, \dots, n_k))$  is recursive, when  $g$  and all  $f_j$  are recursive.
5. Primitive recursion:  $f$  is recursive, where  $f(n_1, \dots, n_k, 0) \simeq g(n_1, \dots, n_k)$ , and for all  $n$ ,  $f(n_1, \dots, n_k, n + 1) \simeq h(n_1, \dots, n_k, n, f(n_1, \dots, n_k, n))$ , where  $g$  and  $h$  are recursive.
6. Minimalisation: suppose that  $g(n_0, n_1, \dots, n_k)$  is a recursive function. Then the partial function  $f(n_1, \dots, n_k)$ , defined to be the value of  $n$  such that  $g(n, n_1, \dots, n_k) = 0$  and for all  $m < n$ ,  $g(m, n_1, \dots, n_k) > 0$ , if this exists, and undefined if it does not, is recursive. We write  $f$  as  $\mu m g(m, n_1, \dots, n_k)$ .

EXAMPLE 3.1.5. we Ackermann's function is recursive but not primitive recursive:

$$\begin{aligned}\psi(0, n) &= n + 1 \\ \psi(m + 1, 0) &= \psi(m, 1)\end{aligned}$$

$$\psi(m+1, n+1) = \psi(m, \psi(m+1, n)).$$

PROOF: (Non-examinable) Ingenious, using induction on the complexity of the demonstration that a function is primitive recursive to show that if  $f(n)$  is primitive recursive, then there exists  $m$  such that the function  $n \mapsto \psi(m, n)$  dominates  $f(n)$  (in the sense that for all but finitely many  $n$ ,  $\psi(m, n) > f(n)$ ); one then deduces that  $n \mapsto \psi(n, n)$  is not primitive recursive and that therefore  $\psi$  isn't either.  $\square$

It grows rather fast.

EMPIRICAL FACT 3.1.6. are (*Church's Thesis*) The recursive partial functions  $f(x_1, \dots, x_k)$  are precisely those that can in principle be calculated by a computer algorithm (that is, such that there is an algorithm that when presented with input  $(a_1, \dots, a_k)$  for which  $f(a_1, \dots, a_k)$  is defined, outputs  $f(a_1, \dots, a_k)$  after a finite time, and when presented with input for  $(a_1, \dots, a_k)$  for which  $f(a_1, \dots, a_k)$  is undefined, runs for ever without halting).

Primitive recursion is expressible in  $\mathcal{L}_E$ .

THEOREM 3.1.7. impressed Every primitive recursive function is provably definable in complexity  $\Sigma_1$  in  $\mathcal{L}_E$  with respect to PAE.

PROOF: We argue by induction on the length of the demonstration that a function is primitive recursive. The only difficult step is when we use primitive recursion. Then we appeal to the following lemma.

LEMMA 3.1.8. and Suppose that  $g : \mathbb{N}^k \rightarrow \mathbb{N}$  and  $h : \mathbb{N}^{k+2} \rightarrow \mathbb{N}$  are functions that can be defined in complexity  $\Sigma_1$ .

Then the function  $f$  defined by primitive recursion from  $g$  and  $h$ , that is to say, defined so that:

1.  $f(n_1, \dots, n_k, 0) = g(n_1, \dots, n_k)$ , and
  2.  $f(n_1, \dots, n_k, n+1) = h(n_1, \dots, n_k, n, f(n_1, \dots, n_k, n))$ ,
- is definable in complexity  $\Sigma_1$ .

PROOF: Consider the statement  $D(y)$ , where  $y$  is a natural number: “ $y$  is the Gödel number of a sequence, the first element of which is  $\overline{[0, g(n_1, \dots, n_k)]}$ , and for each  $m \leq y$ , if  $m$  is a member of the sequence, then for all  $i, j \leq y$  such that  $m = \overline{[i, j]}$ , for all  $m' \leq y$ , if  $m'$  immediately follows  $m$  in the sequence, for all  $i', j' \leq y$  such that  $m' = \overline{[i', j']}$ ,  $i' = i+1$  and  $j' = [i', h(n_1, \dots, n_k, i, j)]$ ”.

This statement is  $\Sigma_1$ , and expresses the idea that  $y$  codes a derivation of values of the function  $f$  using primitive recursion.

We now express “ $z = f(n_1, \dots, n_k, n)$ ” as “There exists  $y$  such that  $D(y)$  holds, and  $[n, z]$  occurs in the sequence coded by  $y$ .”

This is  $\Sigma_1$ .  $\square$

THEOREM 3.1.9. engaged Every recursive partial function is  $\Sigma_1$ -definable, and vice versa.

PROOF:  $\Rightarrow$ ): Easy once we know primitive recursion is expressible. Minimalisation adds an existential quantifier.

$\Leftarrow$ ): Let  $\phi$  be  $\Sigma_0$  such that  $y = f(\mathbf{x}) \leftrightarrow \exists z \phi(\mathbf{x}, y, z)$ .

Roughly speaking, search for  $y$  and  $z$ —or for  $[y, z]$ —such that  $\phi(\mathbf{x}, y, z)$ . If one exists, stop and output  $y$ . If not, return  $\perp$ .

How do we tell if  $\phi(\mathbf{x}, y, z)$ ?

We define primitive recursive  $h_\psi$  which tells whether  $\psi$  is true or not by recursion on  $\Sigma_0$   $\psi$  as follows. We will define  $h_\psi$  to have  $k$  arguments, where  $k$  is largest such that  $v_k$  occurs free in  $\psi$ .

$$h_{v_i=v_j}(n_0, \dots, n_k) = S(1, S(n_i, n_j) + S(n_j, n_i)), \text{ where } k \geq j, i,$$

$$h_{v_i \leq v_j}(n_0, \dots, n_k) = S(1, S(n_i, n_j)) \text{ where } k \geq j, i,$$

$$h_{v_i=\bar{n}}(n_0, \dots, n_i) = S(1, S(n_i, n) + S(n, n_i)),$$

and so on through all the other kinds of atomic formula.

$$h_{\neg\psi}(n_0, \dots, n_k) = S(1, h_\psi(n_0, \dots, n_k)).$$

$h_{\phi \rightarrow \psi}(n_0, \dots, n_k) = S(1, h_\phi(n_1, \dots, n_i).S(1, h_\psi(n_1, \dots, n_j)))$  where  $k$  is the larger of  $i$  and  $j$ .

If  $i, j \leq k$ , then if  $i < j$ , then  $h_{\forall v_i \leq v_j \phi}(n_0, \dots, n_k) = \min_{m \leq n_j} h_\phi(n_0, \dots, n_{i-1}, m, n_{i+1}, \dots, n_{j-1}, n_j)$  with similar definitions in the cases  $i = j$  and  $j < i$ .

$h_{\exists v_i \leq v_j \phi}(n_0, \dots, n_k)$  is defined in a similar way using max instead of min.

Similarly for formulae beginning  $\forall v_i \leq \bar{n}$  and  $\exists v_i \leq \bar{n}$ .

Then express “ $f(\mathbf{x}) = y$ ” as: “ $y$  is the first component of  $[y, z]$ , where  $n = [y, z]$  is least such that  $S(1, h_\phi(\mathbf{x}, y, z)) = 0$ ” (that is, such a pair  $N = [Y, Z]$  exists, and of all the  $N' = [Y', Z'] \leq N$  having the right properties,  $n$  is the least). (Recall that “ $n = [y, z]$ ” is  $\Delta_0$ .)  $\square$

### 3.2. Defining exponentiation

LEMMA 3.2.1. to *The property of being a power of 13 can be defined in  $\mathcal{L}$  in complexity  $\Sigma_0$ .*

PROOF: We express “ $n$  is a power of 13” as: “ $\bar{n} = \bar{1}$ , or  $\bar{13} \mid \bar{n}$ , and if  $p < \bar{n}$  and  $p \mid \bar{n}$  and  $\neg \bar{13} \mid p$ , then  $p = \bar{1}$ ”.  $\square$

LEMMA 3.2.2. fight *The statement “ $n$  is the smallest power of 13 greater than  $m$ ” can be expressed in  $\mathcal{L}$  in complexity  $\Sigma_0$ .*

PROOF: We express it as “ $\bar{n}$  is a power of  $\bar{13}$  and  $\bar{n}$  is greater than  $\bar{m}$  and for all  $k \leq \bar{n}$ , if  $k$  is a power of  $\bar{13}$  greater than  $\bar{m}$ , then  $k = \bar{n}$ ”.  $\square$

LEMMA 3.2.3. forthwith *We can express the concatenation operator  $(m, n) \mapsto m \frown n$  in  $\mathcal{L}$  in complexity  $\Sigma_0$ .*

PROOF: We express “ $k = m \frown n$ ” as “there exists  $l \leq \bar{k}$  such that  $l$  is the smallest power of  $\bar{13}$  greater than  $\bar{n}$ , and  $\bar{k} = \bar{m}.\bar{l} + \bar{n}$ ”.  $\square$

COROLLARY 3.2.4. a *The statement  $k = m^n$ , for  $m, n$  and  $k$  natural numbers, is definable in  $\mathcal{L}$  with respect to PA.*

Moreover, the characteristic function of the graph of exponentiation is  $\Sigma_1$ -definable, so exponentiation is  $\Delta_1$ .

PROOF: Exponentiation is primitive recursive.  $\square$

COROLLARY 3.2.5. power *A function is recursive if and only if it is provably  $\Sigma_1$  in  $\mathcal{L}$  with respect to PA.*

### 3.3. Eliminating exponentiation

PROPOSITION 3.3.1. of Any formula  $\phi$  of  $\mathcal{L}_E$  is provably equivalent from PAE to a formula  $\phi'$  of  $\mathcal{L}$ .

Moreover, for  $n \geq 1$ , if  $\phi$  is  $\Sigma_n$ , then  $\phi'$  can be chosen to be  $\Sigma_n$ , and if  $\phi$  is  $\Pi_n$ , then  $\phi'$  can be chosen to be  $\Pi_n$ .

PROOF: Exponentiation is provably  $\Delta_1$ -definable in  $\mathcal{L}$ .  $\square$

PROPOSITION 3.3.2. english A formula of  $\mathcal{L}$  is provable in PAE if and only if it is provable in PA.

PROOF: Exponentiation can be defined in  $\mathcal{L}$  so that formulae of  $\mathcal{L}$  equivalent to the axioms of PAE can be written in  $\mathcal{L}$ .

In more detail, the axioms of PAE not included in PA fall into two classes.

Firstly, instances of the induction schema written in  $\mathcal{L}_E$ . These are rewritten as equivalent instances of the induction schema written in  $\mathcal{L}$ , which are already axioms of PA.

Secondly, the two statements  $\forall v_1 v_1 \bar{0} = \bar{1}$  and  $\forall v_1 \forall v_2 (v_1^{v_2+} = v_1^{v_2} \cdot v_1)$ . The replacements of these are theorems of PA.  $\square$

COROLLARY 3.3.3. shall Any subset of  $\mathbb{N}^k$ , or any function, that is provably  $\Sigma_n$  or  $\Pi_n$  in PAE is similarly in PA, for  $n \geq 1$ .

### 3.4. Recursive sets

DEFINITION 3.4.1. we A set is recursive if its characteristic function  $\chi_A$  is recursive, and recursively enumerable if the partial function  $\pi_A$  which is 1 on the set and undefined off, is recursive.

THEOREM 3.4.2. levy Equivalently, a set is recursively enumerable iff it is  $\Sigma_1$ , and recursive iff it is  $\Delta_1$ .

PROOF: If  $A$  is recursively enumerable, then  $\pi_A$  is recursive, and hence  $\Sigma_1$ -definable. Then  $A$  is defined by the statement " $\pi_A(\bar{n}) = \bar{1}$ ", which is  $\Sigma_1$ .

Suppose  $A$  is defined by a  $\Sigma_1$  formula  $\phi$ .

Then we define  $\pi_A$  thus: a pair  $(n, m)$  belongs to the graph of  $\pi_A$  if and only if  $\phi(\bar{n})$  and  $\bar{m} = \bar{1}$ . This statement can be expressed in  $\Sigma_1$ .

Suppose  $A$  is recursive. Then  $\pi_A = \pi_{\{1\}} \circ \chi_A$ , which is recursive, and  $\pi_{A^c} = \pi_{\{0\}} \circ \chi_A + 1$ , which is also recursive.

By the above reasoning, both  $A$  and its complement are  $\Sigma_1$ -definable.

Hence  $A$  is  $\Delta_1$ .

Also by the above, if  $A$  is  $\Delta_1$ , then  $A$  and its complement are both  $\Sigma_1$ . Let us suppose that  $A$  is defined by  $\phi$  and its complement by  $\psi$ .

Then we may define  $\chi_A$  in  $\Sigma_1$  as follows: The formula  $\theta(n, m)$  asserting that  $(n, m)$  belongs to the graph of  $\chi_A$  expresses: "either  $\phi(\bar{n})$  and  $\bar{m} = \bar{1}$ , or  $\psi(\bar{n})$  and  $\bar{m} = \bar{0}$ ".

So  $\chi_A$  is  $\Sigma_1$ -definable, and therefore recursive.  $\square$

COROLLARY 3.4.3. whose A set  $A$  is recursive if and only if both  $A$  and its complement are recursively enumerable.

COROLLARY 3.4.4. arms *A subset  $A$  of  $\mathbb{N}$  is recursively enumerable if and only if it is the range of some recursive partial function.*

PROOF: If  $A$  is recursively enumerable, then define  $f(n)$  to be  $n.\pi_A(n)$ . This is recursive, and has range  $A$ .

Now suppose that  $f$  is a recursive partial function with range  $A$ . Since  $f$  is recursive, the statement “ $(n, m)$  is in the graph of  $f$ ” is  $\Sigma_1$ -definable. Now the statement “ $(n, m)$  is in the graph of  $\pi_A$ ” may be expressed as: “there exists  $k$  such that  $(k, \bar{n})$  is in the graph of  $f$ , and  $\bar{m} = \bar{1}$ ”.  $\square$

## 4. Defining provability

### 4.1. Defining expressibility

LEMMA 4.1.1. were “ $n = \ulcorner \bar{m} \urcorner$ ” is expressible in  $\Delta_1$ .

PROOF: We can express it as follows:  $n = 13^m \overline{0} + \sum_{i=0}^{m-1} 13^{i \ulcorner + \urcorner} = 13^m \overline{0}$  since  $\ulcorner + \urcorner = 0$ .  $\square$

COROLLARY 4.1.2. moulded “ $n$  is the Gödel number of a numeral term” is expressible in  $\Delta_1$ .

PROOF: We say: there exists  $m \leq n$  such that  $n = \ulcorner \bar{m} \urcorner$ .  $\square$

LEMMA 4.1.3. in “ $n = \ulcorner v_m \urcorner$ ” is expressible in  $\Delta_1$ .

PROOF: The following are all expressible in  $\Sigma_0$ : the first character of the expression that  $n$  codes for, is  $v$ , and all the other characters are  $'$ ; and  $m$  is largest such that  $13^m \leq n$ .

COROLLARY 4.1.4. their “ $n$  is the Gödel number of some variable term” is expressible in  $\Delta_1$ .

PROOF: We say: there exists  $m \leq n$  such that  $n = \ulcorner v_m \urcorner$ .  $\square$

LEMMA 4.1.5. mothers “ $n$  is the Gödel number of a term” is expressible in  $\Delta_1$ .

PROOF: We argue that there is an algorithm which decides, for any  $n$ , whether or not  $n$  is the Gödel number of a term. It will then follow that the set of Gödel numbers of terms is recursive and therefore  $\Delta_1$ .

The algorithm will proceed as follows (and works by recursion).

If  $n$  is the Gödel number of a numeral term or a variable term, stop, and give the answer “yes”.

If there exists  $m < n$  such that  $m$  is the Gödel number of a term, and  $n = 13.m$ , then stop, and output “yes”.

If there exist  $k, l < n$  such that  $k$  and  $l$  are the Gödel numbers of terms, and  $n$  is the concatenation of  $\ulcorner (, k, \text{any of } \ulcorner f \urcorner, \ulcorner f' \urcorner, \text{or } \ulcorner f'' \urcorner, l, \text{and } \urcorner) \urcorner$ , then stop and output “yes”.

The procedure outlined so far takes only a finite amount of time.

If none of the above are true, then stop and output “no”.  $\square$

LEMMA 4.1.6. womb The notion “ $n$  is the Gödel number of an atomic formula” is expressible in  $\Delta_1$ .

PROOF: We say the following: there exist  $k \leq n$  and  $l \leq n$  such that  $k$  and  $l$  are the Gödel numbers of terms, and  $n$  is the concatenation of  $k$ , one of  $\ulcorner = \urcorner$  or  $\ulcorner \leq \urcorner$ , and  $l$ .  $\square$

LEMMA 4.1.7. to The notion “ $n$  is the Gödel number of a formula” is expressible in  $\Delta_1$ .

PROOF: Similar to the corresponding argument for terms.  $\square$

LEMMA 4.1.8. chase The notion “ $n$  is the Gödel number of a logical axiom” is expressible in  $\Delta_1$ .

PROOF: This follows if there is an algorithm which, for any  $n$ , determines whether or not  $n$  is the Gödel number of a logical axiom.

This is clearly possible. For example, to decide whether  $n$  is the Gödel number of an instance of (A1), all we have to do it to find out whether there exist  $k, l < n$  such that  $k$  and  $l$  are Gödel numbers of formulae, and  $n$  is the concatenation of  $\ulcorner ( \urcorner, k, \ulcorner \rightarrow \urcorner, \ulcorner ( \urcorner, l, \ulcorner \rightarrow \urcorner, k, \urcorner \urcorner$  and  $\urcorner \urcorner$ .  $\square$

## 4.2. Defining proofs and provability

LEMMA 4.2.1. these The statement “ $n$  is the Gödel number of a formula, of which  $m$  is the Gödel number of a proof from assumptions  $S$ ” may be expressed by  $\Delta_i$  formula  $\text{proof}_S(\bar{n}, \bar{m})$ , if  $S$  is definable in  $\Delta_i$  for  $i \geq 1$ .

PROOF: We define  $\text{proof}_S(\bar{n}, \bar{m})$  as follows.

“ $n$  is the Gödel number of a formula.  $m$  codes a sequence whose last member is  $n$ . Whenever  $c$  is a member of this sequence, then either  $c$  codes an axiom, or an assumption, or there exist earlier members  $a$  and  $b$  of the sequence coding formulae which are connected to  $c$  by a rule of inference.”  $\square$

LEMMA 4.2.2. pagans [*provtrue*] Suppose that  $S$  is a set of assumptions definable in  $\Sigma_i$ , where  $i \geq 1$ .

Then there is a  $\Sigma_i$  formula  $\text{Pr}_S(\bar{n})$  which holds in  $\mathbb{N}$  if and only if  $n$  is the Gödel number of a formula provable from  $S$ .

PROOF:  $\text{Pr}_S(\bar{n})$  can be written as  $\exists m \text{proof}_S(\bar{n}, m)$ .

This is  $\Sigma_i$ .  $\square$

LEMMA 4.2.3. in Assume that PAE is consistent, and true in  $\mathbb{N}$ .

If  $S$  is a provably definable set of assumptions, then  $\text{proof}_S(\bar{n}, \bar{m})$  is provable from PAE if and only if  $n$  is the Gödel number of a formula  $\phi$ , and  $m$  is the Gödel number of a proof of  $\phi$  from  $S$ .

Note that we cannot in general express  $\text{Pr}_S(\bar{n})$  in complexity  $\Delta_i$ ; we do have here an increase in complexity.

LEMMA 4.2.4. those Assume that PAE is consistent, and true in  $\mathbb{N}$ .

If  $S$  is a provably definable set of assumptions, then  $\text{Pr}_S(\bar{n})$  is provable from PAE if and only if  $n$  is the Gödel number of a formula provable from  $S$ .

PROOF: Recall that  $\text{Pr}_S(\bar{n})$  can be written as  $\exists m \text{proof}_S(\bar{n}, m)$ .

If  $n = \ulcorner \phi \urcorner$ , where  $\phi$  is provable from  $S$ , then for some  $m$ ,  $\text{proof}_S(\bar{n}, \bar{m})$  is true. If it's true, then we can prove from PAE that  $m$  is the Gödel number of a proof of  $\phi$  from  $S$ .

Hence we can prove that there exists an  $m$  which is the Gödel number of a proof of  $\phi$  from  $S$ ; that is, we can prove  $\text{Pr}_S(\bar{n})$ .

Now suppose that  $\text{Pr}_S(\bar{n})$  is provable from PAE. From our assumption that PAE is true in  $\mathbb{N}$ ,  $\text{Pr}_S(\bar{n})$  is true in  $\mathbb{N}$  by the Soundness Theorem. Hence by Lemma 4.2.3,  $n$  is the Gödel number of a formula provable from  $S$ .  $\square$

**DEFINITION 4.2.5.** holy Suppose that  $S$  is a set of sentences of  $\mathcal{L}$ . Then a proof predicate for  $S$  is a formula  $\text{Pr}_S(x)$  such that for all formulae  $\phi$  of  $\mathcal{L}$ ,  $\phi$  is provable from  $S$  if and only if  $\text{Pr}_S(\ulcorner \phi \urcorner)$  is provable from PAE.

So what we have just proved is that any  $\Delta_i$ -definable set of formulae has a  $\Sigma_i$ -definable proof predicate.

### 4.3. PA and PAE are definable

**THEOREM 4.3.1.** fields PA and PAE are definable in  $\Delta_1$ .

**PROOF:** There is an algorithm deciding whether or not a formula is an element of PA or PAE or not.  $\square$

**COROLLARY 4.3.2.** over  $\Delta_1$  formulae  $\text{proof}_{\text{PA}}(m, n)$  and  $\text{proof}_{\text{PAE}}(m, n)$  expressing that  $n$  codes a proof of the statement coded by  $m$ , in PA and PAE respectively, can be defined, and so can  $\Sigma_1$  proof predicates  $\text{Pr}_{\text{PA}}(n)$  and  $\text{Pr}_{\text{PAE}}(n)$ .

**COROLLARY 4.3.3.** whose If  $n$  is the Gödel number of a formula of  $\mathcal{L}$ , then the statements  $\text{Pr}_{\text{PA}}(\bar{n})$  and  $\text{Pr}_{\text{PAE}}(\bar{n})$  are equivalent.

## 5. Diagonalisation and truth

### 5.1. Diagonalisation

**DEFINITION 5.1.1.** acres Let  $E_n$  be the expression (whatever it is) of Gödel number  $n$ , assuming this exists.

**DEFINITION 5.1.2.** walked  $d(n) = E_n(\bar{n})$ . ( $d$  for “diagonal”.)

**DEFINITION 5.1.3.** those  $D(m, n)$  is the formula  $\bar{n} = \overline{\ulcorner d(m) \urcorner}$ .

**LEMMA 5.1.4.** blessed The statements “ $\bar{m} = d(n)$ ” and  $D(\bar{m}, \bar{n})$  are  $\Delta_1$ .

**THEOREM 5.1.5.** feet (Diagonal Theorem): given a formula  $F(x)$ , there exists a formula  $C$  such that  $C \leftrightarrow F(\overline{\ulcorner C \urcorner})$  is provable in PA.

This is a fixed-point theorem.

**PROOF:** We consider the formula  $F(\overline{\ulcorner d(y) \urcorner})$ .

We define  $\psi(y)$  to be  $F(\overline{\ulcorner d(y) \urcorner})$ .

Let  $k = \ulcorner \psi \urcorner$ .

Let  $C = \psi(\bar{k})$ .

Now  $C$  is  $\psi(\bar{k})$ , which is equivalent to  $F(\overline{\ulcorner d(k) \urcorner})$ .

Also  $k = \ulcorner \psi \urcorner$ , so  $C = E_k[\bar{k}] = d(k)$ . Hence  $F(\overline{\ulcorner d(k) \urcorner}) = F(\overline{\ulcorner C \urcorner})$ .

So  $C$  is equivalent to  $F(\overline{\ulcorner C \urcorner})$ .  $\square$

## 5.2. The undefinability of truth

Truth is not expressible.

**THEOREM 5.2.1.** which (Tarski's Theorem) *There does not exist a formula  $\text{True}(x)$  such that  $\text{True}(\ulcorner\phi\urcorner)$  is true exactly when  $\phi$  is true in  $\mathbb{N}$ .*

**PROOF:** Suppose such a formula to exist.

Then by the Diagonal Lemma, there exists a formula  $C$  such that  $C$  holds if and only if  $\neg \text{True}(\ulcorner C \urcorner)$ .

But then,  $\text{True}(\ulcorner C \urcorner)$  is true if and only if  $C$  is true, if and only if  $\neg \text{True}(\ulcorner C \urcorner)$  is true, giving a contradiction.  $\square$

## 6. Provability

### 6.1. Properties of provability

Provability rules

**THEOREM 6.1.1.** fourteen [provpropone] (The First Provability Rule) *Suppose  $S$  is a provably definable set of assumptions, and  $S \vdash \phi$ . Then  $\text{PA} \vdash \text{Pr}_S(\phi)$ .*

**PROOF:** Write out a formal proof of  $\phi$  from  $S$ . Let  $n$  be its Gödel number. Then  $\text{PA} \vdash \text{proof}_S(\ulcorner\phi\urcorner, \bar{n})$ .

In more detail, there is an algorithm turning a proof  $(\phi_1, \dots, \phi_m)$  of  $\phi$  from the set of assumptions  $S$  into a proof of  $\text{proof}_S(\ulcorner\phi\urcorner, \ulcorner(\phi_1, \dots, \phi_m)\urcorner)$ .

For, let  $\psi_S(x)$  be a formula which provably expresses “ $x$  is the Gödel number of a member of  $S$ ”, that is,  $\psi_S(x)$  is such that  $\text{PA} \vdash \psi_S(\bar{n})$  if  $n$  is the Gödel number of an element of  $S$ , and  $\text{PA} \vdash \neg\psi_S(\bar{n})$  otherwise.

Let  $\psi_{\text{ax}}(x)$  be a formula which provably expresses “ $x$  is the Gödel number of a logical axiom”, let  $\psi_{\text{rule}}(x, y, z)$  express “ $x$  is the Gödel number of a formula which can be obtained by means of a logical rule (MP or Gen) from the formulae with Gödel numbers  $y$  and  $z$ ”. Let  $\psi_{\text{last}}(x, y)$  provably express “ $x$  is a formula, and is the last member of the sequence of formulae whose Gödel number is  $y$ ”.

Then  $\text{proof}_S(x, y)$  expresses the following: “ $x$  is the Gödel number of a formula,  $y$  is the Gödel number of a sequence of formulae, if  $\phi_i$  occurs in this sequence, then  $\psi_S(\ulcorner\phi_i\urcorner)$  or  $\psi_{\text{ax}}(\ulcorner\phi_i\urcorner)$  or for some earlier members  $\phi_j$  and  $\phi_k$ ,  $\psi_{\text{rule}}(\ulcorner\phi_i\urcorner, \ulcorner\phi_j\urcorner, \ulcorner\phi_k\urcorner)$ ; and  $\psi_{\text{last}}(x, y)$ ”.

Now  $\text{proof}_S(\ulcorner\phi\urcorner, \ulcorner(\phi_1, \dots, \phi_m)\urcorner)$  is true, so we can compile a proof of it from PA by putting together the proofs of the various formulae  $\psi_*(x, y)$  that we need.  $\square$

**THEOREM 6.1.2.** hundred [provproptwo] (The Second Provability Rule) *Suppose  $S$  is a definable set of assumptions.  $\text{PA} \vdash \text{Pr}_S(\phi \rightarrow \psi) \rightarrow (\text{Pr}_S \phi \rightarrow \text{Pr}_S \psi)$ .*

**PROOF:** We show that  $\text{PA} \cup \{\text{Pr} \phi, \text{Pr}(\phi \rightarrow \psi)\} \models \text{Pr} \psi$ , and deduce the result from that.

Suppose that in a model of PA,  $n_1$  and  $n_2$  are elements of the model and the statements  $\text{proof}(\ulcorner\phi\urcorner, n_1)$  and  $\text{proof}(\ulcorner(\phi \rightarrow \psi)\urcorner, n_2)$  are true in that model.

Let  $\hat{n}_1$  be the initial part of  $n_1$  which codes a sequence omitting the last  $\ulcorner\#\urcorner$  from the sequence coded by  $n_1$ .

Then if  $n = \hat{n}_1 \hat{\ } n_2 \ulcorner\psi\urcorner \ulcorner\#\urcorner$ , then  $\text{proof}(\ulcorner\psi\urcorner, n)$  holds in that model.  $\square$



**THEOREM 6.1.3.** years [provpropthree] (*The Third Provability Rule*) Suppose that  $S$  is a provably definable set of assumptions, including PA as a subset.

Then  $\text{PA} \vdash \text{Pr}_S(\overline{\psi}) \rightarrow \text{Pr}_S(\overline{\text{Pr}_S(\overline{\psi})})$ .

**PROOF:** This is an arithmetised version of the proof of Theorem 6.1.1.

Consider the statement “ $m$  is the Gödel number of a proof of  $\phi$  from  $S$  which is  $l$  steps long”; write as  $\text{proof}_S^*(\overline{\phi}, m, l)$ . Note that  $\text{proof}_S(\overline{\phi}, m)$  is equivalent to  $\exists l \text{proof}_S^*(\overline{\phi}, m, l)$ . We will argue inductively that for all  $l$ , if  $\text{proof}_S^*(\overline{\phi}, m, l)$  holds, then  $\text{Pr}_S(\overline{\text{Pr}_S(\overline{\phi})})$  holds; the argument will proceed by recursively constructing the Gödel number  $M$  of a proof in  $S$  of  $\text{Pr}_S(\overline{\phi})$ , and noting that  $\text{proof}_S(\overline{\text{Pr}_S(\overline{\phi})}, M)$  holds.

We will do this in a general model of PA, which means we need to be careful, because in arbitrary models of PA,  $\text{Pr}_S(\overline{\psi})$  does not necessarily entail  $S \vdash \psi$ .

Suppose that  $\mathfrak{N}$  is a model of PA, and that  $\mathfrak{N} \models \text{Pr}_S(\overline{\phi})$ .

Then there exist  $m, l \in \mathfrak{N}$  such that  $\mathfrak{N} \models \text{proof}_S^*(\overline{\phi}, m, l)$ . In the argument that follows we need to bear in mind that  $\mathfrak{N}$  may not be  $\mathbb{N}$ , and that  $m$  and  $l$  may not be actual natural numbers; they just have, in  $\mathfrak{N}$ , some of the first-order properties that natural numbers possess.

We argue using induction on  $l$  (this can be formalised in PA, using the induction scheme). We examine the inductive step; the base case is similar, but easier.

Suppose then that in  $\mathfrak{N}$ ,  $l > 1$ , and that  $\mathfrak{N} \models \text{proof}_S^*(i, m, l)$ . Suppose, using the inductive hypothesis, that if, in  $\mathfrak{N}$ ,  $j$  is the Gödel number other than the last one of an element of the sequence whose Gödel number is  $m$ , then there exists  $M_j$  in  $\mathfrak{N}$  such that  $\mathfrak{N} \models \text{proof}_S(\overline{\text{Pr}_S(i)}, M_j)$ .

If in  $\mathfrak{N}$ ,  $i$  is the Gödel number of a member of  $S$ , that is to say, if  $\mathfrak{N} \models \psi_S(i)$ , then we let  $M_j$  be the Gödel number of a proof of  $\psi_S(i)$ .

To justify this step further, recall that in  $\mathbb{N}$ ,  $\psi_S(\overline{n})$  is provable if  $n$  is the Gödel number of a member of  $S$ , and  $\neg\psi_S(\overline{n})$  is provable if not. We shall argue that there is an algorithm which, when presented with input  $n$ , outputs the Gödel number of a proof of  $\psi_S(\overline{n})$  if  $n \in S$ , and outputs the Gödel number of a proof of  $\neg\psi_S(\overline{n})$  if not. The algorithm goes like this. Examine all formal proofs in  $S$ , one by one. (They can be listed in a recursive way that permits us to do this). We will eventually encounter either a proof of  $\psi_S(\overline{n})$ , in which case we output its Gödel number; or we encounter a proof of  $\neg\psi_S(\overline{n})$ , in which case we output the Gödel number of that. The existence of this algorithm means that there is a  $\Sigma_1$ -definable function  $f_S$  inputting  $n$  and outputting the Gödel number of the appropriate proof. Suppose that  $\chi(x, y)$  expresses “ $y = f_S(x)$ ”. Then if we are in the situation of the previous paragraph, and  $\mathfrak{N} \models \psi_S(i)$ , then there exists  $M_i \in \mathfrak{N}$  such that  $\mathfrak{N} \models \chi(i, M_i)$ . Construct  $M_i$  by appending  $N_i$  and a proof of  $\psi_{\text{last}}$ .

In a similar way, if in  $\mathfrak{N}$ ,  $i$  is the Gödel number of a logical axiom, then let  $M_i$  be the Gödel number of a proof of  $\psi_{\text{ax}}(i)$ .

If  $i$  is, in  $\mathfrak{N}$ , the Gödel number of a formula obtained using an application of a rule to formulae earlier in the sequence coded by  $n$  whose Gödel numbers are  $j$  and  $k$ , then by the inductive hypothesis there exist elements  $M_j$  and  $M_k$  of  $\mathfrak{N}$  such that  $\mathfrak{N} \models \text{proof}_S(j, M_j)$  and  $\mathfrak{N} \models \text{proof}_S(k, M_k)$ . Then we generate  $M_i$  by combining  $M_j$  and  $M_k$ , deleting repeated

#'s as necessary.  $\square$

Add somewhere: PA is provably definable.

## 6.2. A limit on the power of proof

**THEOREM 6.2.1.** ago (Weak form of the First Incompleteness Theorem) *Suppose that  $S$  is a provably definable set of sentences that is true in  $\mathbb{N}$  and includes PA.*

*Then there exists a formula  $G$  such that  $G$  is true in  $\mathbb{N}$ , but is not provable from  $S$ .*

**PROOF:** Using the Diagonal Lemma, find a formula  $G$  such that  $G \leftrightarrow \neg \text{Pr}_S(\ulcorner G \urcorner)$  is provable from PA.

Suppose that  $G$  is provable from  $S$ . Then  $G$  is true in  $\mathbb{N}$ , by assumption and by soundness. Also  $\text{Pr}_S(\ulcorner G \urcorner)$  is true by Lemma 4.2.2. But then  $G$  is false, contradiction.

So  $G$  is not provable from  $S$ .

Now suppose that  $G$  is false. Then  $\text{Pr}_S(\ulcorner G \urcorner)$  is true, so  $G$  is provable from  $S$  by Lemma 4.2.2. But  $S$  is true in  $\mathbb{N}$ , so  $G$  is true in  $\mathbb{N}$  also by soundness, contradiction.

So  $G$  is true in  $\mathbb{N}$ .  $\square$

## 6.3. Grades of completeness

**DEFINITION 6.3.1.** were *A set of axioms  $S$  is  $n$ -inconsistent if and only if there exists a  $\Sigma_n$  formula  $\exists x \phi(x)$  such that  $\vdash_S \exists x \phi(x)$ , but for all  $m$ ,  $\vdash_S \neg \phi(\overline{m})$ .*

*$S$  is  $n$ -consistent if and only if it is not  $n$ -inconsistent.*

*$S$  is  $\omega$ -consistent if and only if it is  $n$ -consistent for all  $n$ .*

**DEFINITION 6.3.2.** nailed *A set  $S$  of statements is  $\Sigma_i$ -complete if and only if all true  $\Sigma_i$ -sentences are provable from  $S$ .*

*We will say it is  $\Sigma_i$ -sound if and only if all  $\Sigma_i$ -sentences provable from  $S$  are true.*

**DEFINITION 6.3.3.** for *The axiom scheme  $Q$  is the following list of axioms:*

- $\forall v_1 \forall v_2 v_1^+ = v_2^+ \rightarrow v_1 = v_2.$
- $\forall v_1 \neg v_1^+ = \overline{0}.$
- $\forall v_1 v_1 + \overline{0} = v_1.$
- $\forall v_1 \forall v_2 v_1 + v_2^+ = (v_1 + v_2)^+.$
- $\forall v_1 v_1 \cdot \overline{0} = \overline{0}.$
- $\forall v_1 \forall v_2 v_1 \cdot v_2^+ = v_1 \cdot v_2 + v_1.$
- $\forall v_1 v_1 \leq \overline{0} \leftrightarrow v_1 = \overline{0}.$
- $\forall v_1 \forall v_2 v_1 \leq v_2^+ \leftrightarrow (v_1 \leq v_2 \vee v_1 = v_2^+).$
- $\forall v_1 \forall v_2 v_1 \leq v_2 \vee v_2 \leq v_1.$

This has no induction.

**DEFINITION 6.3.4.** our *The following list of axioms is known as  $R$ .*

- All sentences  $\overline{m} + \overline{n} = \overline{k}$ , for which  $m + n = k$ .*
- All sentences  $\overline{m} \cdot \overline{n} = \overline{k}$ , for which  $m \cdot n = k$ .*
- All sentences  $\overline{m} \neq \overline{n}$ , where  $m \neq n$ .*
- All sentences  $\forall v_1 v_1 \leq \overline{n} \leftrightarrow (v_1 = \overline{0} \vee \dots \vee x = \overline{n})$ .*
- All sentences  $\forall v_1 v_1 \leq \overline{n} \vee \overline{n} \leq v_1$ .*

**PROPOSITION 6.3.5.** advantage  *$Q$  extends  $R$ .*

THEOREM 6.3.6. to  $R$  is  $\Sigma_0$ -complete.

PROOF: The second-to-last schema gives a method of eliminating bounded quantifiers.

The other axioms allow us to compute the diagrams of  $+$ ,  $\cdot$  and  $\leq$ .  $\square$

COROLLARY 6.3.7. the  $Q$  and  $PA$  are  $\Sigma_0$ -complete.

THEOREM 6.3.8. bitter Any system  $S$  that is  $\Sigma_0$ -complete is also  $\Sigma_1$ -complete.

PROOF: Suppose that  $\exists v_1 F(v_1)$  is true. Then  $F(\bar{n})$  is true for some  $n$ . Then  $S \vdash F(\bar{n})$  by  $\Sigma_0$ -completeness. So  $S \vdash \exists v_1 F(v_1)$  as required.  $\square$

COROLLARY 6.3.9. cross  $R$ ,  $Q$ , and  $PA$  are  $\Sigma_1$ -complete.

Presburger arithmetic is  $PA$  with all mention of multiplication erased.

DEFINITION 6.3.10. but The following list of statements, in the sublanguage  $\mathcal{L}_P$  of  $\mathcal{L}$  containing no uses of the multiplication symbol  $f'$ , is known as Presburger arithmetic:

1.  $\neg \forall v_i v_i^+ = \bar{0}; \forall v_i \forall v_j (v_i^+ = v_j^+ \rightarrow v_i = v_j)$ .  
( $n \mapsto n^+$  is an injection from  $\mathbb{N} \leftrightarrow \mathbb{N} \setminus \{0\}$ ).
2.  $\forall v_i v_i + \bar{0} = v_i$ .
3.  $\forall v_i \forall v_j v_i + v_j^+ = (m + n)^+$ .
4.  $\forall v_i \bar{0} \leq v_i; \forall v_i \forall v_j (v_i \leq v_j \leftrightarrow (v_i = v_j \vee v_i^+ \leq v_j)); \forall v_i v_i \leq v_i; \forall v_i \forall v_j (v_i \leq v_j \vee v_j \leq v_i); \forall v_i \forall v_j \forall v_k ((v_i \leq v_j \wedge v_j \leq v_k) \rightarrow (v_i \leq v_k)); \forall v_i \forall v_j ((v_i \leq v_j \wedge v_j \leq v_i) \rightarrow v_i = v_j)$ .  
( $\leq$  is a total order, with initial element  $0$ , and  $n^+$  is the immediate successor of  $n$ ).
5. (Induction Schema): For any formula  $\phi(v_1)$  of  $\mathcal{L}_P$ , the following is an axiom: if  $\phi(0)$ , and if for all  $n$ ,  $\phi(n)$  implies  $\phi(n^+)$ , then  $\forall n \phi(n)$ .

Formally:

$$((\phi(\bar{0}) \wedge (\forall v_1 \phi(v_1) \rightarrow \phi(v_1^+))) \rightarrow \forall v_1 \phi(v_1)).$$

The following theorem is not examinable for part C or OMMS.

THEOREM 6.3.11. this Presburger arithmetic is consistent and complete, and the set of consequences of it is decidable.

PROOF: Rather long, very ingenious, and involving quantifier elimination and modular arithmetic.  $\square$

So multiplication is not definable in  $\mathcal{L}_P$ .

## 6.4. The first incompleteness theorem

THEOREM 6.4.1. our (First Incompleteness Theorem) There exists a  $\Pi_1$ -sentence  $G$  such that if  $PA$  is consistent, then  $PA \not\vdash G$ , and if in addition  $PA$  is 1-consistent, then  $PA \not\vdash \neg G$ .

PROOF: Apply the Diagonal Lemma to the statement  $\neg \text{Pr}_{PA}(\ulcorner v_1 \urcorner)$ , to obtain a formula  $G$  such that  $G$  is provably equivalent to  $\neg \text{Pr}_{PA}(\ulcorner G \urcorner)$ .

Suppose that  $PA \vdash G$ .

Then  $PA \vdash \text{Pr}_{PA}(\ulcorner G \urcorner)$ , by Theorem 6.1.1.

So since  $PA \vdash G$ ,  $PA \vdash \neg \text{Pr}_{PA}(\ulcorner G \urcorner)$ .

Thus  $PA$  is inconsistent, giving a contradiction.

Now suppose that  $PA \vdash \neg G$ , and that PA is 1-consistent.

Then  $PA \vdash \text{Pr}_{PA}(\overline{\neg G})$ , because  $G$  is provably equivalent to  $\neg \text{Pr}_{PA}(\overline{\neg G})$ .

Now  $\text{Pr}_{PA}(\overline{\neg G})$  is the same thing as  $\exists x \text{ proof}_{PA}(\overline{\neg G}, x)$ , and  $\text{proof}_{PA}(\overline{\neg G}, x)$  is  $\Sigma_1$ .

Write  $\text{proof}_{PA}(\overline{\neg G}, x)$  as  $\exists y \phi(x, y)$ , where  $\phi(x, y)$  is  $\Sigma_0$ .

Then  $\exists x \exists y \phi(x, y)$  may be rewritten  $\exists z \exists x \leq z \exists y \leq z \phi(x, y)$ , which is  $\Sigma_1$  in the strict sense.

Now  $PA \vdash \exists z \exists x \leq z \exists y \leq z \phi(x, y)$ , so because PA is 1-consistent, there must exist  $n$  such that  $PA \not\vdash \neg \exists x \leq \bar{n} \exists y \leq \bar{n} \phi(x, y)$ .

But PA is  $\Sigma_0$ -complete by Lemma 5.3.6., so  $PA \vdash \exists x \leq \bar{n} \exists y \leq \bar{n} \phi(x, y)$ .

So this statement is true in  $\mathbb{N}$ . Let  $m$  be such that  $\mathbb{N} \models \exists y \leq \bar{n} \phi(\bar{m}, y)$ .

Then  $\mathbb{N} \models \text{proof}_{PA}(\overline{\neg G}, \bar{m})$ .

Hence  $m$  is the Gödel number of a proof of  $G$  in PA.

So we can read off a proof of  $G$  in PA from  $m$ , and see that  $PA \vdash G$ .

Hence PA is inconsistent, giving a contradiction.  $\square$

We can generalise this to other axiom systems which imply PA, except that we weaken the conclusion so that  $G$  becomes  $\Pi_n$  where  $n \geq 1$  and the given axiom system is  $\Delta_n$ .

**COROLLARY 6.4.2.** *purpose* Assume  $\mathbb{N}$  is a model of PA. Then  $G$  is true in  $\mathbb{N}$  and not provable.

**PROOF:** If  $\mathbb{N}$  is a model of PA, then PA is consistent and 1-consistent.

Hence PA proves neither  $G$  nor  $\neg G$ .

Because PA is  $\Sigma_1$ -complete, and because PA does not prove  $\neg G$ ,  $\neg G$  must be false in  $\mathbb{N}$ , so  $G$  is true in  $\mathbb{N}$ .  $\square$

**THEOREM 6.4.3.** *now* (*Rosser's Theorem*) Let  $S$  be any provably definable consistent set of sentences including PA. Then there is a sentence  $G$  such that  $S$  neither proves nor disproves  $G$ .

**PROOF:** Let  $H(x)$  be the statement  $\exists y (\text{proof}_S(\overline{\neg \neg x}, y) \wedge \forall z \leq y \neg \text{proof}_S(x, z))$ .

(Informally,  $H(\overline{\neg \phi})$  says “there is a  $y$  coding a refutation of  $\phi$ , and no  $z \leq y$  codes a proof of  $\phi$ ”.)

Using the Diagonal Lemma, let  $G$  be such that  $G \leftrightarrow H(\overline{\neg G})$  is provable from PA.

We argue that  $S$  neither proves nor refutes  $G$ .

Suppose first that  $S \vdash G$ .

Then there is a proof of  $G$  from  $S$ . Let  $n$  be its Gödel number.

Then  $PA \vdash \text{proof}_S(\overline{\neg G}, \bar{n})$ , and so  $S \vdash \text{proof}_S(\overline{\neg G}, \bar{n})$ .

Now  $S$  is consistent, so given that  $S \vdash G$ , then it is not the case that  $S \vdash \neg G$ ; and so no disproof of  $G$  exists.

So no natural number  $m$  is the Gödel number of a proof from  $S$  of  $\neg G$ ; in particular no natural number  $m < n$  is the Gödel number of a proof from  $S$  of  $\neg G$ .

So if  $m < n$ ,  $PA \vdash \neg \text{proof}_S(\overline{\neg G}, m)$ , so that  $PA \vdash \forall m < \bar{n} \neg \text{proof}_S(\overline{\neg G}, m)$ .

Now  $n$  is the Gödel number of a proof of  $G$  from  $S$ , so  $PA \vdash \forall m \geq n \exists x \leq m \text{ proof}_S(\overline{\neg G}, x)$  (the value of  $x$  that witnesses this is of course  $n$  itself).

Putting these two sentences together,  $PA \vdash \forall y \neg (\text{proof}_S(\overline{\neg G}, y) \wedge \forall z \leq y \neg \text{proof}_S(\overline{\neg G}, z))$ .

That is,  $PA \vdash \neg H(\overline{\neg G})$ . Hence  $S \vdash \neg H(\overline{\neg G})$ .

But  $S \vdash G$ , so  $S \vdash H(\overline{\neg G})$ , giving a contradiction.

Now suppose that  $S \vdash \neg G$ .

Then there is a proof of  $\neg G$  from  $S$ . Let  $n$  be the Gödel number of that proof.

Since  $S$  is consistent, it is not possible that  $S \vdash G$ . So there is no proof of  $G$  from  $S$ .

So for all  $m \leq n$ ,  $m$  is not the Gödel number of a proof of  $G$  from  $S$ .

Hence  $\text{PA} \vdash (\text{proof}(\overline{\neg G}), \overline{n}) \wedge \forall m \leq n \neg \text{proof}_S(G, m)$ .

So  $S$  proves the same thing.

Hence  $S \vdash H(\overline{\neg G})$ . From this it follows that  $S \vdash G$ , giving a contradiction.  $\square$

This theorem really doesn't use all of PA: just that you have the Diagonal Lemma and you can identify all genuine natural numbers as occurring near the beginning.

## 6.5. The Second Incompleteness Theorem and Löb's Theorem

**THEOREM 6.5.1.** is (Second Incompleteness Theorem) *If  $S$  is a provably definable set of sentences including PA, and if a sentence  $G$  has the property that  $S \vdash G \leftrightarrow \neg \text{Pr}_S(\overline{G})$ , then  $S \vdash \neg \text{Pr}_S(\overline{X}) \rightarrow \neg \text{Pr}_S(\overline{G})$ .*

**PROOF:**  $(G \rightarrow (\neg G \rightarrow X))$  is a tautology and so a theorem of  $S$ .

By hypothesis,  $S \vdash (\text{Pr}_S(\overline{G}) \rightarrow \neg G)$ .

Hence  $S \vdash (G \rightarrow (\text{Pr}_S(\overline{G}) \rightarrow X))$ .

From Theorem 6.1.1, and the assumption that  $S$  extends PA, it follows that  $S \vdash \text{Pr}_S(\overline{(G \rightarrow (\text{Pr}_S(\overline{G}) \rightarrow X))})$ .

From Theorem 6.1.2, we have  $S \vdash (\text{Pr}_S(\overline{(G \rightarrow (\text{Pr}_S(\overline{G}) \rightarrow X))}) \rightarrow (\text{Pr}_S(\overline{G}) \rightarrow \text{Pr}_S(\overline{(\text{Pr}_S(\overline{G}) \rightarrow X)})$ .

Hence  $S \vdash (\text{Pr}_S(\overline{G}) \rightarrow \text{Pr}_S(\overline{(\text{Pr}_S(\overline{G}) \rightarrow X)}))$ .

Also from Theorem 6.1.2, we have  $S \vdash (\text{Pr}_S(\overline{(\text{Pr}_S(\overline{G}) \rightarrow X)}) \rightarrow (\text{Pr}_S(\overline{\text{Pr}_S(\overline{G})}) \rightarrow \text{Pr}_S(\overline{X}))$ .

Thus  $S \vdash (\text{Pr}_S(\overline{G}) \rightarrow (\text{Pr}_S(\overline{\text{Pr}_S(\overline{G})}) \rightarrow \text{Pr}_S(\overline{X})))$ .

Now by Theorem 6.1.3, we have that  $S \vdash (\text{Pr}_S(\overline{G}) \rightarrow \text{Pr}_S(\overline{\text{Pr}_S(\overline{G})}))$ .

Thus  $S \vdash (\text{Pr}_S(\overline{G}) \rightarrow \text{Pr}_S(\overline{X}))$ .

Hence  $S \vdash (\neg \text{Pr}_S(\overline{X}) \rightarrow \neg \text{Pr}_S(\overline{G}))$ , as required.  $\square$

**COROLLARY 6.5.2.** twelve *If  $S$  is a provably definable set of sentences including PA, and if a sentence  $G$  has the property that  $S \vdash G \leftrightarrow \neg \text{Pr}_S(\overline{G})$ ,  $X$  is a sentence, and  $S$  is consistent, then  $S \not\vdash \neg \text{Pr}(\overline{X})$ .*

**PROOF:** If  $G$  exists, and  $S \vdash \neg \text{Pr}(\overline{X})$ , then  $S \vdash \neg \text{Pr}(\overline{G})$ , so  $S \vdash G$ . But then  $S \vdash \text{Pr}(\overline{G})$ . So  $S$  is inconsistent.  $\square$

**DEFINITION 6.5.3.** month *Suppose that  $S$  is a definable set of sentences. We define  $\text{Con}_S$  to be the formula  $\neg \text{Pr}_S(\overline{\neg 0 = 0})$ . We read this as “ $S$  is consistent”.*

**COROLLARY 6.5.4.** old *If  $S$  is a provably definable set of sentences including PA, and  $S$  is consistent, then it is not the case that  $S \vdash \text{Con}_S$ .*

**PROOF:** In fact,  $S$  does not prove the statement  $\neg \text{Pr}_S(\overline{X})$  for any formula  $X$ .  $\square$

**THEOREM 6.5.5.** and (Löb's Theorem) If the Diagonal Lemma is provable in  $S$ , and  $P$  is a provability predicate for  $S$ , then from  $S \vdash (P(\overline{\neg\phi}) \rightarrow \phi)$  we can deduce  $S \vdash \phi$ . Suppose that  $S$  is a provably definable set of sentences extending  $PA$ . Then from  $S \vdash (\text{Pr}_S(\overline{\neg\phi}) \rightarrow \phi)$  we can deduce  $S \vdash \phi$ .

**PROOF:** Let  $L$  be diagonal for  $\text{Pr}_S(\cdot) \rightarrow \phi$ , ie  $S \vdash (L \leftrightarrow (\text{Pr}_S(\overline{L}) \rightarrow \phi))$ .

Then by Theorem 6.1.1,  $S \vdash \text{Pr}_S(\overline{L \rightarrow (\text{Pr}_S(\overline{L}) \rightarrow \phi)})$ .

By Theorem 6.1.2,  $S \vdash \text{Pr}_S(\overline{L}) \rightarrow \text{Pr}_S(\text{Pr}_S(\overline{L}) \rightarrow \phi)$ .

By Theorem 6.1.2,  $S \vdash \text{Pr}_S(\overline{L}) \rightarrow (\text{Pr}_S(\overline{\text{Pr}_S(\overline{L})}) \rightarrow \text{Pr}_S(\overline{\neg\phi}))$ , so  $S \vdash ((\text{Pr}_S(\overline{L}) \rightarrow \text{Pr}_S(\overline{\text{Pr}_S(\overline{L})})) \rightarrow (\text{Pr}_S(\overline{L}) \rightarrow \text{Pr}_S(\overline{\neg\phi})))$  by (A2), so  $S \vdash (\text{Pr}_S(\overline{L}) \rightarrow \text{Pr}_S(\overline{\neg\phi}))$  by Theorem 6.1.3 and MP.

Using HS,  $S \vdash \text{Pr}_S(\overline{L}) \rightarrow \phi$ .

But this is equivalent to  $L$ , so  $S \vdash L$ .

By Theorem 6.1.1,  $S \vdash \text{Pr}_S(\overline{L})$ .

Now by MP,  $S \vdash \phi$  as required.  $\square$

## 6.6. A stronger version of $\Sigma_1$ -completeness

Tidy this up and make it a bit more rigorous.

We proved earlier that if  $\phi$  is  $\Sigma_1$  and true, then it is provable.

In this section we strengthen this result.

**THEOREM 6.6.1.** bootless If  $\phi$  is a  $\Sigma_1$  sentence, then  $PA \vdash (\phi \rightarrow \text{Pr}_{PA}(\overline{\neg\phi}))$ .

**SKETCH PROOF:** We show that if  $\mathfrak{N}$  is a model of  $PA$ , then  $\mathfrak{N} \models \phi \rightarrow \text{Pr}_{PA}(\overline{\neg\phi})$ .

So, assume that  $\mathfrak{N} \models PA$ .

Then we carry out, in  $\mathfrak{N}$ , the construction of a partial function  $F$  such that, if  $\phi$  is a  $\Sigma_0$  sentence, then

$$\mathfrak{N} \models \phi \rightarrow \underset{PA}{\text{proof}}(\overline{\neg\phi}, F(\overline{\neg\phi}) \wedge \overline{\neg\phi} \wedge \overline{\neg\#}),$$

and

$$\mathfrak{N} \models \neg\phi \rightarrow \underset{PA}{\text{proof}}(\overline{\neg\neg\phi}, F(\overline{\neg\phi}) \wedge \overline{\neg\neg\phi} \wedge \overline{\neg\#}).$$

We do this by a messy induction on  $\phi$ , of which the messiest part is when  $\phi$  is atomic or negated atomic.

The following special cases can be done algorithmically using induction:

1.  $\overline{n} = \overline{n}$  is a logical axiom.
2.  $\neg\overline{m} = \overline{n}$  where  $m \neq n$ .
3.  $\overline{m} + \overline{n} = \overline{m+n}$  and  $\overline{m} \cdot \overline{n} = \overline{m \cdot n}$ .

Bounded quantifiers can be coped with as follows.

If  $\phi = \forall m \leq n \psi(m)$ , then the following procedure can be expressed in  $\mathcal{L}$ : for each  $m \leq n$ , write down a proof of  $\psi(\overline{m})$ , deduce  $\bigwedge_{m \leq n} \psi(\overline{m})$ , and then by induction on  $n$  deduce  $\forall m \leq n \psi(\overline{m})$ ; the result is a proof of  $\phi$ .

We treat bounded existential quantifiers in a similar way.

Now  $\overline{m} \leq \overline{n}$  is provably equivalent to  $\exists k \leq \overline{n} \overline{m} + k = \overline{n}$ .

Now any  $\Sigma_0$  formula is provably equivalent, by standard proofs that can be expressed in  $\mathcal{L}$ , to a disjunction of conjunctions of statements of the above forms.

So, for any  $\Sigma_0$  formula  $\psi$ , we have

$$\mathfrak{N} \models \psi \rightarrow \text{proof}_{\text{PA}}(\overline{\ulcorner \psi \urcorner}, F(\overline{\ulcorner \psi \urcorner}) \wedge \overline{\ulcorner \psi \urcorner} \wedge \overline{\ulcorner \# \urcorner}),$$

and

$$\mathfrak{N} \models \neg\psi \rightarrow \text{proof}_{\text{PA}}(\overline{\ulcorner \neg\psi \urcorner}, F(\overline{\ulcorner \psi \urcorner}) \wedge \overline{\ulcorner \neg\psi \urcorner} \wedge \overline{\ulcorner \# \urcorner}).$$

Now suppose  $\phi$  is a  $\Sigma_1$  formula  $\exists x \psi(x)$ .

Then

$$\mathfrak{N} \models \forall x \left( \psi(x) \rightarrow \text{proof}_{\text{PA}}(\overline{\ulcorner \psi(x) \urcorner}, F(\overline{\ulcorner \psi(x) \urcorner}) \wedge \overline{\ulcorner \psi(x) \urcorner} \wedge \overline{\ulcorner \# \urcorner}) \right),$$

so

$$\mathfrak{N} \models \forall x \left( \psi(x) \rightarrow \text{Pr}_{\text{PA}}(\overline{\ulcorner \psi(x) \urcorner}) \right),$$

so

$$\mathfrak{N} \models \exists x (\psi(x)) \rightarrow \exists x \text{Pr}_{\text{PA}}(\overline{\ulcorner \psi(x) \urcorner}),$$

so

$$\mathfrak{N} \models \exists x (\psi(x)) \rightarrow \text{Pr}_{\text{PA}}(\overline{\ulcorner \exists x \psi(x) \urcorner}),$$

as required.  $\square$

**THEOREM 6.6.2.** tis If  $\phi(x)$  is  $\Sigma_1$ , then the statement  $\forall x (\phi(x) \rightarrow \text{Pr}_{\text{PA}}(\overline{\ulcorner \phi(x) \urcorner}))$  is a theorem of PA.

**SKETCH PROOF:** This is proved by the same techniques as the previous theorem.  $\square$

## 7. Strengthenings of PA

Given that PA is incomplete, we look around for reasonable strengthenings of it. We could use  $\text{PA} \cup \text{Con}_{\text{PA}}$ ,  $\text{PA} \cup \text{Con}_{\text{PA}} \cup \text{Con}_{\text{PA} \cup \text{Con}_{\text{PA}}}$ , etc. The next section provides a more systematic possible approach.

### 7.1. The $\omega$ -rule

**DEFINITION 7.1.1.** to Suppose that  $S$  is a set of formulae of  $\mathcal{L}$ .

We define  $S^\omega$  to be the logical system whose axioms are  $S$  together with all logical axioms, and whose rules are MP, Gen, and the  $\omega$ -rule which allows one to deduce  $\forall x \phi(x)$  from the entire set of assumptions  $\{\phi(\overline{n}) : n \in \mathbb{N}\}$ .

A proof in  $S^\omega$  is a sequence  $(\phi_\alpha : \alpha < \beta)$ , where  $\beta$  is an ordinal, such that each  $\phi_\alpha$  is an element of  $S$  or a logical axiom, or else is obtained from previous members of the sequence using a rule.

$\phi$  is a theorem of  $S^\omega$ , and we write  $S^\omega \vdash \phi$ , iff there is a proof in  $S^\omega$  of which  $\phi$  is the last element.

We could, if we wished, insist that all proofs have length  $< \omega_1$ .

The  $\omega$ -rule looks reasonable-ish. However there is a big problem with it.

**THEOREM 7.1.2.** tell  $R^\omega$  is complete.

PROOF: We can prove by induction on the complexity of a formula  $\phi$  that  $R^\omega \vdash \phi$  or  $R^\omega \vdash \neg\phi$ .

The  $\omega$ -rule allows us to eliminate quantifiers.

The case where  $\phi$  is  $\Sigma_0$  is already done since  $R$  is  $\Sigma_0$ -complete.

Now suppose that  $\phi$  is  $\Sigma_{n+1}$ ; say  $\phi = \exists x \psi(x)$ .

There are two cases. If there exists  $n$  such that  $R^\omega \vdash \psi(\bar{n})$ , then it is certainly true that  $R^\omega \vdash \exists x \psi(x)$ , so  $R^\omega \vdash \phi$ . The alternative is (appealing to the inductive hypothesis) that for all  $n$ ,  $R^\omega \vdash \neg\psi(\bar{n})$ . Then by the  $\omega$ -rule,  $R^\omega \vdash \forall x \neg\psi(x)$ , so  $R^\omega \vdash \neg\phi$ .

This argument of course also does the case when  $\phi$  is  $\Pi_{n+1}$ .  $\square$

COROLLARY 7.1.3.  $\boxed{\text{you}}$   $\text{PA}^\omega$  is complete.

COROLLARY 7.1.4.  $\boxed{\text{we}}$  Assuming that  $R^\omega$  and  $\text{PA}^\omega$  are sound with respect to truth in  $\mathbb{N}$ , then the set of theorems of  $R^\omega$  or of  $\text{PA}^\omega$  is undefinable and so a fortiori not recursively enumerable.

PROOF: Using Tarski's Theorem, and the statement that a set is recursively enumerable iff it is  $\Sigma_1$ -definable.  $\square$

The upshot is that since, as human beings, we are limited to what is recursively enumerable,  $R^\omega$  and  $\text{PA}^\omega$  are of no practical use.

In the next section we look at an adaptation of the  $\omega$ -rule which may be more useful.

## 7.2. The uniform reflection principle

The uniform reflection principle is an arithmetised version of the  $\omega$ -rule, and says “if  $\phi(\bar{n})$  is provable for all  $n$ , then  $\forall x \phi(x)$  is true” (which can be said in the language).

DEFINITION 7.2.1.  $\boxed{\text{will}}$  The uniform reflection principle *URP* is the set of axioms got by adding to  $\text{PA}$  all instances of the following, where  $F(v_1)$  is a formula of  $\mathcal{L}$ :

$$\forall n \text{Pr}_{\text{PA}}(\overline{\ulcorner \forall v_1 (v_1 = \ulcorner \neg \neg \bar{n} \neg \neg \rightarrow F(v_1) \urcorner \rceil})} \rightarrow \forall n F(n)).$$

We write this as  $\forall n \text{Pr}_{\text{PA}}(\overline{\ulcorner F[\bar{n}] \urceil}) \rightarrow \forall n F(n)$ , and refer to it as the reflection principle for  $F$ .

This is better—we have a definable set of axioms here—so less powerful. How powerful?

THEOREM 7.2.2.  $\boxed{\text{go}}$  Suppose that  $G$  is a sentence such that  $\text{PA} \vdash G \leftrightarrow \neg \text{Pr}_{\text{PA}}(\overline{\ulcorner G \urceil})$ .

Then  $\text{PA} \vdash \forall n \text{Pr}_{\text{PA}}(\overline{\ulcorner \neg \text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, \dot{n}) \urceil})$ .

PROOF: Recall that  $\text{proof}_{\text{PA}}$  is  $\Delta_1$ .

Suppose that  $\mathfrak{N}$  is a model of  $\text{PA}$ , and that  $n \in \mathfrak{N}$ .

Then either  $\text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, n)$  is true in  $\mathfrak{N}$ , or  $\neg \text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, n)$  is true.

If  $\mathfrak{N} \models \neg \text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, n)$  is true, then because  $\neg \text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, n)$  is  $\Sigma_1$ , then by Theorem 6.6.2.,  $\mathfrak{N} \models \forall x (\neg \text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, x) \rightarrow \text{Pr}_{\text{PA}}(\neg \text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, x)))$ , so we deduce that  $\mathfrak{N} \models \text{Pr}_{\text{PA}}(\overline{\ulcorner \neg \text{proof}_{\text{PA}}(\overline{\ulcorner G \urceil}, \bar{n}) \urceil})$ .



If on the other hand  $\text{proof}_{\text{PA}}(\overline{\neg G}, n)$ , then  $\text{Pr}_{\text{PA}}(\overline{\neg G})$ , and so  $\text{Pr}_{\text{PA}}(\overline{\neg X})$  for all  $X$  by the Second Incompleteness Theorem, and so  $\text{Pr}_{\text{PA}}(\overline{\neg \neg \text{proof}_{\text{PA}}(\overline{\neg G}, n)})$  in particular.  $\square$

COROLLARY 7.2.3. therefor  $\text{URP} \vdash G$ .

PROOF: Using the previous theorem and  $\text{URP}$ , we have  $\text{URP} \vdash \forall n \neg \text{proof}_{\text{PA}}(\overline{\neg G}, n)$ , that is,  $\text{URP} \vdash \neg \text{Pr}(\overline{\neg G})$ , from which we deduce  $\text{URP} \vdash G$ .  $\square$

So  $\text{URP}$  is stronger than  $\text{PA}$ . By how much?

THEOREM 7.2.4. we Writing  $\text{URP}_{\Pi_1}$  for the axiom system got by adding to  $\text{PA}$  only instances of the reflection principle for  $\Pi_1$  formulae,  $\text{PA} \cup \text{URP}_{\Pi_1}$  is equivalent to  $\text{PA} \cup \{\text{Con}_{\text{PA}}\}$ .

PROOF: Assume  $\text{PA} \cup \text{URP}_{\Pi_1}$ . We set out to prove  $\text{Con}_{\text{PA}}$ .

Now  $\text{Con}_{\text{PA}}$  is  $\Pi_1$ , so express it as  $\forall y \psi(y)$ , where  $\psi(y)$  is  $\Sigma_0$ .

Then  $\psi(y)$  is  $\Sigma_1$ , so we can prove from  $\text{PA}$  that  $\forall y (\psi(y) \rightarrow \text{Pr}_{\text{PA}}(\overline{\neg \psi(y)}))$ .

But we can also prove  $\neg \psi(y) \rightarrow \neg \forall z \psi(z)$ , which is to say,  $\neg \psi(y) \rightarrow \neg \text{Con}_{\text{PA}}$ . Now  $\neg \text{Con}_{\text{PA}}$  is the statement  $\text{Pr}_{\text{PA}}(\overline{\neg \perp})$ , where  $\perp$  is a contradiction, and since  $\perp \rightarrow \chi$  is provable for any  $\chi$ , in particular  $\perp \rightarrow \psi(y)$  is provable; and now, by using the First and Second Provability Rules,  $\text{Pr}_{\text{PA}}(\overline{\neg \perp}) \rightarrow \text{Pr}_{\text{PA}}(\overline{\neg \psi(y)})$  is provable. Thus we have  $\neg \psi(y) \rightarrow \text{Pr}_{\text{PA}}(\overline{\neg \psi(y)})$ . We can indeed prove  $\forall y (\neg \psi(y) \rightarrow \text{Pr}_{\text{PA}}(\overline{\neg \psi(y)}))$ .

Thus we can prove  $\forall y \text{Pr}_{\text{PA}}(\overline{\neg \psi(y)})$ . Now  $\text{URP}_{\Pi_1}$  gives us  $\forall y \psi(y)$ , as required.

Now assume  $\text{Con}_{\text{PA}}$  and  $\forall z \text{Pr}_{\text{PA}}(\overline{\neg \psi(z)})$ , for some  $\Sigma_0$ -formula  $\psi(z)$ .

Then we must have  $\forall z \neg \text{Pr}_{\text{PA}}(\overline{\neg \psi(z)})$ , because  $\neg \psi(z) \rightarrow (\psi(z) \rightarrow \perp)$  is a theorem, for any contradiction  $\perp$ , and so, using the First and Second Provability Rules, so is  $\text{Pr}_{\text{PA}}(\overline{\neg \psi(z)}) \rightarrow (\text{Pr}_{\text{PA}}(\overline{\neg \psi(z)}) \rightarrow \text{Pr}_{\text{PA}}(\overline{\neg \perp}))$ . But  $\text{Pr}_{\text{PA}}(\overline{\neg \perp})$  is false by hypothesis.

But now, since  $\neg \psi(z)$  is  $\Sigma_0$  and therefore  $\Sigma_1$ , we can prove  $\forall z (\neg \psi(z) \rightarrow \text{Pr}(\neg \psi(z)))$ .

Thus  $\forall z \psi(z)$  follows, as required.  $\square$

Any  $\Pi_1$ -theorem of  $\text{PA}$  is a theorem of  $\text{PA}$  with induction only for  $\Pi_2$  plus  $\text{Con}_{\text{PA}}$ : omit this?

If  $A$  and  $B$  are disjoint and recursively enumerable, then  $A$  and  $B$  can be recursively separated by a  $\Sigma_1$  formula.

## 8. Gödel-Löb logic

Here we abstract out some of the features of the logic of provability we've been deriving, finding that a surprisingly small part of it is sufficient to give us the Incompleteness Theorems.

### 8.1. Definitions and basic results

DEFINITION 8.1.1. meet Gödel-Löb logic is a system of modal propositional logic.

The symbols are: a countably infinite number of propositional variables  $p, q, r$  etc; a logical constant  $\perp$ , a binary connective  $\rightarrow$ , and a unary operator  $\square$ .

The formulae are: all propositional variable letters; the symbol  $\perp$ ; and all strings  $(\phi \rightarrow \psi)$  and  $\square \phi$  where  $\phi$  and  $\psi$  are formulae.

The logical axioms are all propositional tautologies (with  $\perp$  interpreted as a contradiction), together with all instances of  $\Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$ , and  $\Box(\Box\phi \rightarrow \phi) \rightarrow \Box\phi$ . The rules of inference are modus ponens and necessitation, by which we mean the rule “if  $\vdash \phi$ , then  $\vdash \Box\phi$ ”.

$\Box\phi$  is to be interpreted “ $\phi$  is provable”.

LEMMA 8.1.2. not If  $A$  and  $B$  are formulae of GL logic, then  $\vdash \Box(A \wedge B) \leftrightarrow (\Box A \wedge \Box B)$ .

PROOF: Assume  $\Box(A \wedge B)$ .

Now  $(A \wedge B) \rightarrow A$  is a propositional tautology.

Using Necessitation,  $\Box((A \wedge B) \rightarrow A)$  is a theorem.

Now using the second axiom scheme and MP, we have  $\Box(A \wedge B) \rightarrow \Box A$ .

Similarly we have  $\Box(A \wedge B) \rightarrow \Box B$ , so  $\vdash \Box(A \wedge B) \rightarrow (\Box A \wedge \Box B)$ .

Now assume  $\Box A \wedge \Box B$ .

Now  $A \rightarrow (B \rightarrow (A \wedge B))$  is a propositional tautology.

Using Necessitation, the second axiom scheme, and MP, we have  $\Box A \rightarrow (\Box B \rightarrow \Box(A \wedge B))$ , which is equivalent to  $(\Box A \wedge \Box B) \rightarrow \Box(A \wedge B)$ .  $\square$

THEOREM 8.1.3. now For all formulae  $\phi$ ,  $\Box X \rightarrow \Box\Box X$  is a theorem of GL logic.

PROOF: The formula  $X \rightarrow ((\Box\Box X \wedge \Box X) \rightarrow (\Box X \wedge X))$  is a propositional tautology.

So since  $\vdash \Box(\Box X \wedge X) \rightarrow (\Box\Box X \wedge \Box X)$ , we have that  $\vdash X \rightarrow (\Box(\Box X \wedge X) \rightarrow (\Box X \wedge X))$ .

So by Necessitation,  $\vdash \Box(X \rightarrow (\Box(\Box X \wedge X) \rightarrow (\Box X \wedge X)))$ .

Hence, using the second axiom schema and Modus Ponens,  $\vdash \Box X \rightarrow \Box(\Box(\Box X \wedge X) \rightarrow (\Box X \wedge X))$ .

But  $\Box(\Box(\Box X \wedge X) \rightarrow (\Box X \wedge X)) \rightarrow \Box(\Box X \wedge X)$  is an axiom.

So we have that  $\vdash \Box X \rightarrow \Box(\Box X \wedge X)$ .

Now by the previous lemma,  $\vdash \Box(\Box X \wedge X) \rightarrow (\Box\Box X \wedge \Box X)$ .

Hence  $\vdash \Box X \rightarrow (\Box\Box X \wedge \Box X)$ .

So  $\vdash \Box X \rightarrow \Box\Box X$ , as required.  $\square$

The following theorem shows that the abstraction process is very successful.

THEOREM 8.1.4. then (Solovay Completeness Theorem) Suppose that  $\phi$  is a formula of GL logic.

Then  $\vdash \phi$  if and only if whenever  $\psi \mapsto \psi^*$  is a map from formulae of GL logic to formulae of  $\mathcal{L}$  having the properties that  $\perp^*$  is a contradiction, that  $(\psi \rightarrow \chi)^* = (\psi^* \rightarrow \chi^*)$ , and that  $(\Box\psi)^* = \text{Pr}_{\text{PA}}(\ulcorner \psi^* \urcorner)$ ,  $\text{PA} \vdash \phi^*$ .

PROOF: The forward direction is relatively easy. The reverse direction involves clever use of what are known as *Kripke frames*. We will do this later in the course.  $\square$

The following feature of propositional logic carries over.

PROPOSITION 8.1.5. let (Substitution) Suppose that  $\phi$ ,  $\chi$ ,  $\psi$  and  $\theta$  are formulae of Gödel-Löb logic, and that  $\theta'$  is obtained from  $\theta$  by replacing one or more subformulae of  $\theta$  that are copies of  $\chi$ , by copies of  $\psi$ .

Then  $\phi \rightarrow (\chi \leftrightarrow \psi) \vdash (\phi \rightarrow (\theta \leftrightarrow \theta'))$ .

PROOF: Induction on the complexity of  $\theta$ .  $\square$

PROPOSITION 8.1.6. me (Modalised substitution) Suppose that  $X = X(p)$  is a formula in which  $p$  only occurs within the scope of  $\square$  operators, and let  $X(q)$  be the result of replacing all instances of  $p$  in  $X$  by  $q$ .

Then  $\vdash (\square(p \leftrightarrow q) \rightarrow (X(p) \leftrightarrow X(q)))$ .

PROOF: Induction on the complexity of  $X$ .  $\square$

## 8.2. The fixed-point theorem for GL logic

Fixed point theorem; more abstract proof of incompleteness.

THEOREM 8.2.1. hear Fixed point theorem: if  $A(p)$  is a formula in which  $p$  only occurs in the scope of a  $\square$ , then there exists a formula  $X$ , in which  $p$  does not occur and containing only letters from  $A(\cdot)$ , such that  $X \leftrightarrow A(X)$  is provable.

Moreover,  $X$  is unique in the sense that  $\vdash ((\square(p \leftrightarrow A(p)) \wedge \square(q \leftrightarrow A(q))) \rightarrow \square(p \leftrightarrow q))$ .

LEMMA 8.2.2. of If  $B(p)$  is a formula, then there exists a formula  $X$ , in which  $p$  does not occur and containing only letters from  $B(\cdot)$ , such that  $X \leftrightarrow \square B(X)$  is provable.

PROOF: The appropriate  $X$  is:  $\square B(\top)$ , where  $\top$  is some tautology (such as  $\perp \rightarrow \perp$ ).

For,  $\square B(\top) \rightarrow (\top \leftrightarrow \square B(\top))$  is a tautology.

Thus, using substitution, so is  $\square B(\top) \rightarrow (\square B(\top) \leftrightarrow \square B(\square B(\top)))$ .

So we get  $\square B(\top) \rightarrow \square B(\square B(\top))$ .

As for the other way round, given  $\square B(\top) \rightarrow (\top \leftrightarrow \square B(\top))$ , we use substitution again to get  $\square B(\top) \rightarrow (B(\top) \leftrightarrow B(\square B(\top)))$ .

It follows by propositional logic that  $B(\square B(\top)) \rightarrow (\square B(\top) \rightarrow B(\top))$ .

By necessitation,  $\square(B(\square B(\top)) \rightarrow (\square B(\top) \rightarrow B(\top)))$ .

Using the first axiom and MP,  $\square B(\square B(\top)) \rightarrow \square(\square B(\top) \rightarrow B(\top))$ .

The second axiom scheme gives us  $\square(\square B(\top) \rightarrow B(\top)) \rightarrow \square B(\top)$ .

Now by propositional logic,  $\square B(\square B(\top)) \rightarrow \square B(\top)$  as required.  $\square$

LEMMA 8.2.3. you Given a set of formulae  $C_i(D(p_1, \dots, p_n))$  ( $i \leq n$ ), there exist formulae  $F_i$  for  $i \leq n$  such that  $\vdash (F_i \leftrightarrow \square C_i(D(F_1, \dots, F_n)))$ .

PROOF: We do induction on  $n$ .

The base case was done above. Suppose that any such family of equivalences of size  $n$  can be solved, and suppose that we have a family of formulae  $C_i(D(p_1, \dots, p_{n+1}))$  ( $i \leq n+1$ ).

Then we set  $C_{n+1}$  aside for a moment. Let  $q$  be some propositional letter we have not yet used. Using the inductive hypothesis, let  $G_i(q)$  (for  $i \leq n$ ) be formulae such that

$$\vdash G_i(q) \leftrightarrow \square C_i(D(G_1(q), \dots, G_n(q), q))$$

( $i \leq n$ ).

Now use the preceding lemma to find  $F_{n+1}$  such that

$$\vdash F_{n+1} \leftrightarrow \square C_{n+1}(D(G_1(F_{n+1}), \dots, G_n(F_{n+1}), F_{n+1})).$$

Now, for  $i \leq n$ , let  $F_i = G_i(F_{n+1})$ .  $\square$

LEMMA 8.2.4. *my* (*Existence of the fixed point*) If  $A(p)$  is a formula in which  $p$  only occurs in the scope of a  $\square$ , then there exists a formula  $X$ , in which  $p$  does not occur and containing only letters from  $A(\cdot)$ , such that  $X \leftrightarrow A(X)$  is provable.

PROOF: Suppose that  $A(p)$  has the form  $D(\square C_1(p), \dots, \square C_n(p))$ . Use the preceding lemma to find  $F_i$  equivalent to  $\square C_i(D(F_1, \dots, F_n))$  for  $i \leq n$ ; then  $D(F_1, \dots, F_n)$  is equivalent to  $D(\square C_1(D(F_1, \dots, F_n)), \dots, \square C_n(D(F_1, \dots, F_n)))$ , that is, to  $A(D(F_1, \dots, F_n))$ , which is what we want.  $\square$

LEMMA 8.2.5. *gentle* (*Uniqueness of the fixed point*) Suppose that  $A(p)$  is a formula in which  $p$  only occurs in the scope of a  $\square$ , and that  $X$  is a formula in which  $p$  does not occur and containing only letters from  $A(\cdot)$ , such that  $X \leftrightarrow A(X)$  is provable.

Then  $X$  is unique in the sense that  $\vdash ((\square(p \leftrightarrow A(p)) \wedge \square(q \leftrightarrow A(q))) \rightarrow \square(p \leftrightarrow q))$ .

PROOF: We prove, using modalised substitution, that  $\vdash \square(p \leftrightarrow q) \rightarrow (A(p) \leftrightarrow A(q))$ .

By propositional logic,  $\vdash ((p \leftrightarrow A(p)) \wedge (q \leftrightarrow A(q))) \rightarrow (\square(p \leftrightarrow q) \rightarrow (p \leftrightarrow q))$ .

Doing stuff with  $\square$ , we get  $\vdash ((\square(p \leftrightarrow A(p)) \wedge \square(q \leftrightarrow A(q))) \rightarrow \square(\square(p \leftrightarrow q) \rightarrow (p \leftrightarrow q)))$ .

Then, using an axiom,  $\vdash (\square(p \leftrightarrow A(p)) \wedge \square(q \leftrightarrow A(q))) \rightarrow \square(p \leftrightarrow q)$ .  $\square$

As an example of silly things happen if  $p$  is not boxed, let  $A$  be the identity.

### 8.3. The incompleteness theorems in GL logic

THEOREM 8.3.1. *cousin* (GL version of the First Incompleteness Theorem). *There exists a formula  $G$  such that  $\vdash (G \leftrightarrow \neg \square G)$ .*

PROOF: Define  $G$  to be a/the fixed point of  $\neg \square p$ ; that is,  $\vdash G \leftrightarrow \neg \square G$ .  $\square$

THEOREM 8.3.2. *westmoreland* (GL version of the Second Incompleteness Theorem). *For any formulae  $A$  and  $B$ ,  $\vdash \square \neg \square A \rightarrow \square B$ .*

PROOF: Now  $\vdash (\neg \square A \rightarrow (\square A \rightarrow A))$ , so  $\vdash (\square \neg \square A \rightarrow \square(\square A \rightarrow A))$ , so  $\vdash (\square \neg \square A \rightarrow \square A)$ . So since, by Theorem 8.1.2.,  $\vdash (\square A \rightarrow \square \square A)$ ,  $\vdash (\square \neg \square A \rightarrow \square \square A)$ .

Now  $(\neg \square A \rightarrow (\square A \rightarrow B))$  by propositional calculus.

Hence, using Necessitation, the scheme  $(\square(\phi \rightarrow \psi) \rightarrow (\square \phi \rightarrow \square \psi))$ , and MP,

$$\vdash (\square \neg \square A \rightarrow (\square \square A \rightarrow \square B)).$$

Then using propositional calculus,

$$\vdash (\square \neg \square A \rightarrow \square B)$$

as required.  $\square$

The formula  $\square \neg \square A \rightarrow \square B$  expresses the idea that if anything is provably unprovable, then the system is inconsistent.

## 8.4. Kripke semantics

DEFINITION 8.4.1. what A Kripke frame is a triple  $\mathfrak{K} = (W, R, v)$ , where  $W$  is a non-empty set of worlds,  $R$  is a relation on  $W$ , and  $v$  is a function such that for each  $w \in W$  and each propositional letter  $p$ ,  $v(w, p) \in \{T, F\}$ .

DEFINITION 8.4.2. yesternight If  $\phi$  is a formula of GL logic,  $\mathfrak{K} = (W, R, v)$  is a Kripke frame, and  $w \in W$ , then we say that  $\phi$  is true at  $w$ , and write  $\mathfrak{K}, w \models \phi$ , under the following circumstances:

1. It is never the case that  $\mathfrak{K}, w \models \perp$ .
  2.  $\mathfrak{K}, w \models p$ , where  $p$  is a propositional letter, if and only if  $v(w, p) = T$ .
  3.  $\mathfrak{K}, w \models (\chi \rightarrow \psi)$  if and only if either  $\mathfrak{K}, w \not\models \chi$ , or  $\mathfrak{K}, w \models \psi$ .
  4.  $\mathfrak{K}, w \models \Box \psi$  if and only if for all  $w'$  such that  $w R w'$ ,  $\mathfrak{K}, w' \models \psi$ .
- If  $\mathfrak{K}, w \models \phi$  for all  $w \in W$ , we write  $\mathfrak{K} \models \phi$ , and say that  $\phi$  is true in  $\mathfrak{K}$ .

THEOREM 8.4.3. our Let  $\phi$  be a formula of GL logic. Then  $\phi$  is a theorem if and only if it is true in every Kripke frame in which the accessibility relation is a strict partial order with no infinite ascending chains.

We prove this in two parts.

PROPOSITION 8.4.4. council If  $\phi$  is a theorem of GL logic, then it is true in every Kripke frame in which the accessibility relation is irreflexive and transitive and has no infinite ascending chains.

PROOF: We argue that all axioms are true in all such frames, and that the set of formulae true in all such frames is closed under application of the rules of inference.

Every propositional tautology is certainly true in every frame.

The truth of  $\Box(\phi \rightarrow \psi) \rightarrow (\Box \phi \rightarrow \Box \psi)$  is an easy exercise.

Now we verify  $\Box(\Box \phi \rightarrow \phi) \rightarrow \Box \phi$ .

Suppose that  $\mathfrak{K}, w \models \Box(\Box \phi \rightarrow \phi)$ , but it is not the case that  $\mathfrak{K}, w \models \Box \phi$ .

Then there exists some  $w'$  such that  $w R w'$  and  $\mathfrak{K}, w' \models \neg \phi$ .

However  $\mathfrak{K}, w \models \Box(\Box \phi \rightarrow \phi)$ , so  $\mathfrak{K}, w' \models \Box \phi \rightarrow \phi$ . So  $\mathfrak{K}, w' \models \neg \Box \phi$ .

Let  $w_1 = w'$ . Given  $w_n$  such that  $w R w_n$  and  $w_n \models \neg \phi$ , we deduce as in the previous paragraph that  $\mathfrak{K}, w_n \models \neg \Box \phi$ . Thus we can find  $w_{n+1}$  such that  $w_n R w_{n+1}$  and  $\mathfrak{K}, w_{n+1} \models \neg \phi$ . Since  $R$  is transitive,  $w R w_{n+1}$ , and the process continues.

Thus  $R$  admits an infinite ascending chain, contrary to our assumption.

So all axioms are true in all frames satisfying the given conditions.

Now we confirm that the formulae true in all frames, are closed under the rules of inference.

This is clear for Modus Ponens.

As for Necessitation, if  $\phi$  is true in all frames  $\mathfrak{K}$  of the appropriate kind, then for all  $w$ ,  $\mathfrak{K}, w \models \phi$  holds, and so we must have that for all  $w$ ,  $\mathfrak{K}, w \models \Box \phi$ , so  $\mathfrak{K} \models \Box \phi$ .  $\square$

Note that the truth of  $\Box X \rightarrow \Box \Box X$  follows at once from transitivity of  $R$ .

PROPOSITION 8.4.5. did We argue that if  $\phi$  is not a theorem of GL logic, then there is a Kripke frame  $(W, R, v)$ , where  $W$  is finite, and  $R$  is irreflexive and transitive, that does not affirm it.

PROOF: Let  $S$  be the set of all formulae of the forms  $\psi$ ,  $\neg\psi$ ,  $\Box\psi$  or  $\neg\Box\psi$  for  $\psi$  a subformula of  $\phi$ .

Let  $W$  be the set of all subsets  $w$  of  $S$  such that

$$\theta_w = \bigwedge_{\psi \in w} \psi \wedge \bigwedge_{\psi \in S \setminus w} \neg\psi$$

is consistent, and if  $w, w' \in W$ , then we say that  $w R w'$  if and only if

1. If  $\Box\psi \in w$ , then  $\psi, \Box\psi \in w'$ ,
2.  $F(w) = w \cap \{\Box\psi : \psi \in S\}$  is a strict subset of  $F(w')$ .

Clearly  $(W, R)$  is a finite strict partial order.

Transitivity: suppose  $w R w' R w''$ .

Condition 2. clearly works.

We now look at condition 1.

Suppose  $\Box\psi \in w$ .

Then  $\Box\psi \in w'$ , so  $\psi, \Box\psi \in w''$ , as required.

For a propositional letter  $p$ , if  $p \in S$ , then say  $p$  is true at  $w$  if and only if  $p \in w$ ; otherwise say  $p$  is true at  $w$ .

We check that  $\theta_w$  is true at  $w$ .

We check, for each  $\psi \in S$ , that  $\psi$  is true at  $w$  if and only if  $\psi \in w$  by induction on  $\psi$ , as follows.

For  $\psi$  a propositional letter or  $\perp$ , it is obvious. It is also clear if  $\psi$  had the form  $\psi_1 \rightarrow \psi_2$ .

Now suppose that  $\psi = \Box\chi$ .

If  $\psi \in w$ , then by definition of  $R$ ,  $\chi$  must belong to  $w'$  whenever  $w R w'$ , so since by the inductive hypothesis  $\chi$  is true at all such  $w'$ ,  $\psi$  is true at  $w$ .

If  $\psi \notin w$ , then  $\{\Box\eta : \Box\eta \in w\} \cup \{\neg\Box\chi\}$  is consistent, so so is  $\{\Box\eta : \Box\eta \in w\} \cup \{\neg(\Box\chi \rightarrow \chi)\}$ , since  $\neg\Box\chi \rightarrow \neg\Box(\Box\chi \rightarrow \chi)$  is a theorem of GL logic.

Hence  $\{\eta, \Box\eta : \Box\eta \in w\} \cup \{\Box\chi, \neg\chi\}$  is also consistent.

Extend it to a complete consistent set, and let  $w'$  be the intersection of that complete consistent set intersection with  $S$ .

Then  $w' \in W$ , and  $w R w'$ , and so  $\psi$  is false at  $w$  as required.

Having done that, we extend  $\{\neg\phi\}$  to a complete consistent set, and let  $w$  be the intersection of that complete consistent set with  $S$ .

Then  $\phi$  is false at  $w$ , as required.  $\square$

## 9. Constructing models of PA inside other models of PA

### 9.1. Revision of Henkin's proof of the Completeness Theorem

Recall how the now standard proof of the Completeness Theorem goes. Given a countable language  $L$  of first-order predicate calculus, we close the language under the addition of constant symbols  $c_\phi$  for formulae  $\phi$ , and add to PA (or whatever other theory we may be interested in) extra axioms  $(\exists v_1 \phi \rightarrow \phi(c_\phi))$ . We then extend our theory to a complete consistent set of sentences, and a new model for the theory is then constructed from the closed terms (sc. terms containing no variable letters), in such a way that if

$\phi(\sigma_1, \dots, \sigma_n)$  is a member of the complete consistent set, then  $\phi(\sigma_1, \dots, \sigma_n)$  is true in that model.

So, once we have a process for deciding which sentences are members of our complete consistent set, the construction of the model is routine.

We focus on this process of identifying members of the complete consistent set.

First, we add the constant symbols to our language  $\mathcal{L}$ . We can do this in any number of ways. The method in the following definition relies on the fact that the string  $\bar{0}$  never occurs in formulae of  $\mathcal{L}$ .

**DEFINITION 9.1.1.** decree We define a language  $\mathcal{L}^*$  with the same alphabet as  $\mathcal{L}$ , with its set of terms and its set of formulae being the smallest sets which have the following properties.

- (1) Every term of  $\mathcal{L}$  is a term of  $\mathcal{L}^*$ , and every formula of  $\mathcal{L}$  is a formula of  $\mathcal{L}^*$ .
- (2) If  $\phi$  is a formula of  $\mathcal{L}^*$ , then  $\bar{0}'(\phi)$  is a term of  $\mathcal{L}^*$  (which we will write as  $c_\phi$ ).
- (3) If  $\sigma$  and  $\tau$  are terms of  $\mathcal{L}^*$ , then so are  $\sigma^+$ ,  $(\sigma \text{ f } \tau)$  and  $(\sigma \text{ f' } \tau)$ .
- (4) If  $\sigma$  and  $\tau$  are terms of  $\mathcal{L}^*$ , then  $\sigma = \tau$  and  $\sigma \leq \tau$  are formulae of  $\mathcal{L}^*$ .
- (5) If  $\phi$  and  $\psi$  are formulae of  $\mathcal{L}^*$ , then so are  $\neg\phi$ ,  $(\phi \rightarrow \psi)$ , and  $\forall v_i \phi$ .

We now add witnesses to existential statements to any theory of  $\mathcal{L}$ .

**DEFINITION 9.1.2.** in If  $T$  is any set of sentences of  $\mathcal{L}$ , then we define  $T^*$  to be the result of adding to  $T$  all formulae  $(\exists v_1 \phi(v_1) \rightarrow \phi[c_\phi])$ .

**THEOREM 9.1.3.** forwarding If  $T$  is  $\Sigma_n$ -definable for  $n \geq 1$ , then  $T^*$  is also  $\Sigma_n$ -definable. Similarly if  $T$  is  $\Pi_n$ -definable for  $n \geq 1$ , then  $T^*$  is also  $\Pi_n$ -definable.

**PROOF:** The set of Gödel numbers of the extra axioms in  $T^*$  is  $\Delta_1$ -definable.  $\square$

We now take a closer look at the process of deriving the complete consistent set.

**THEOREM 9.1.4.** this Suppose that  $T$  has a  $\Delta_n$ -definable proof predicate  $\text{Pr}_T$ . Then there is a  $\Delta_n$ -definable function  $\mathfrak{H}_T(n)$  such that if  $n$  is the Gödel number of a sentence  $\phi$  of  $\mathcal{L}^*$ , then in a model  $\mathfrak{N}$  of PA,  $\mathfrak{H}_T(\phi) = 1$  if, defining  $\theta_n$  to be the formula

$$\theta_n = \bigwedge \{ \psi : \ulcorner \psi \urcorner < n \wedge \mathfrak{H}_T(\ulcorner \psi \urcorner) = 1 \} \wedge \bigwedge \{ \neg \psi : \ulcorner \psi \urcorner < n \wedge \mathfrak{H}_T(\ulcorner \psi \urcorner) = 0 \},$$

$\text{Pr}_T(\overline{\ulcorner \theta_n \rightarrow \phi \urcorner})$  is true, and  $\mathfrak{H}_T(n) = 0$  otherwise.

We describe  $\mathfrak{H}_T$  as successful if there exists  $n$  such that  $\mathfrak{H}_T(n) = 0$ .

Note that  $\mathfrak{H}_T$  is successful in a model  $\mathfrak{N}$  of PA if and only if  $\mathfrak{N} \models \text{Con}_T$ .

**PROOF:** To see that  $\mathfrak{H}_T$  is  $\Delta_n$ -definable, note that the definition of  $\mathfrak{H}_T$  is a definition by recursion using a  $\Delta_n$ -formula.  $\square$

If  $\mathfrak{H}_T$  is successful in a model  $\mathfrak{N}$ , then we think of  $\mathfrak{H}_T$  as describing a model of  $T$  inside  $\mathfrak{N}$ . It is certainly the case that from a successful function  $\mathfrak{H}_T$ , we can construct a model of  $T$ .

## 9.2. Comparing a model and a model inside that model

When we start to talk about models of arithmetic sitting inside models of arithmetic, questions about language and metalanguage, and problems posed by non-standard elements coding formulae and proofs, become complicated. For some results in this section, we restrict our attention to models of arithmetic sitting inside  $\mathbb{N}$ .

**THEOREM 9.2.1.** dear *If  $\mathfrak{N}$  is a model of PA defined in  $\mathbb{N}$  by a formula  $\mathfrak{H}_T$ , where  $T$  extends PA, then  $\mathfrak{N}$  is not elementarily equivalent to  $\mathbb{N}$  (that is, there is a sentence which is true in  $\mathfrak{N}$  and false in  $\mathbb{N}$ ).*

**PROOF:** This follows from Tarski's Theorem. If  $\mathfrak{N}$  and  $\mathbb{N}$  were elementarily equivalent, then  $\mathfrak{H}_T$  would define truth in  $\mathbb{N}$ , which is impossible.  $\square$

This theorem can be extended, with extreme caution, to other models of PA.

**THEOREM 9.2.2.** expedience *If  $\phi$  is a sentence of  $\mathcal{L}$ , and  $\mathfrak{N}$  is a model of PA, then there is a model  $\mathfrak{N}'$  of PA inside  $\mathfrak{N}$  satisfying  $\phi$  if and only if  $\mathfrak{N} \models \neg \text{Pr}_{\text{PA}}(\ulcorner \neg \phi \urcorner)$ , and every model  $\mathfrak{N}'$  of PA inside  $\mathfrak{N}$  satisfies  $\phi$  if and only if  $\mathfrak{N} \models \text{Pr}_{\text{PA}}(\ulcorner \phi \urcorner)$ .*

Restricting our attention to  $\mathfrak{H}_{\text{PA}}$ , we have the following.

**THEOREM 9.2.3.** my *There is a  $\Delta_2$ -sentence  $K$  such that if  $\mathfrak{N}$  is a model of PA and  $\mathfrak{N}'$  is constructed inside  $\mathfrak{N}$  using  $\mathfrak{H}_{\text{PA}}$ , then  $K$  is true in  $\mathfrak{N}$  if and only if it is false in  $\mathfrak{N}'$ .*

**PROOF:** Use the Diagonal Lemma to find  $K$  such that  $\text{PA} \vdash (K \leftrightarrow (\neg \mathfrak{H}_{\text{PA}}(\ulcorner K \urcorner))) = \bar{1}$ .

We can regard  $K$  as being  $\Delta_2$  because  $\mathfrak{H}_{\text{PA}}$  is.  $\square$

**COROLLARY 9.2.4.** liege *Any chain  $\mathfrak{N}_i$  of models of PA, where  $\mathfrak{N}_{i+1}$  is constructed inside  $\mathfrak{N}_i$  using  $\mathfrak{H}_{\text{PA}}$ , is finite.*

**PROOF:** We order functions from  $\mathbb{N}$  to  $\{0, 1\}$  lexicographically, that is, so that  $f \leq g$  iff either  $f = g$ , or there exists  $n$  such that for all  $m < n$ ,  $f(m) = g(m)$ , and  $f(n) < g(n)$ .

We define  $f_i$  so that for all  $n \in \mathbb{N}$ ,  $f_i(n) = 1$  if and only if  $n$  is the Gödel number of a sentence  $\phi$  such that  $\mathfrak{N}_i \models \phi$ .

Let  $k = \ulcorner K \urcorner$ . Then for all  $i$ ,  $f_i(k) \neq f_{i+1}(k)$ .

We now note that if  $i > 1$ , then  $f_i \leq f_{i+1}$ . For, suppose that  $n$  is least such that  $f_i(n) \neq f_{i+1}(n)$ . Then in one of  $\mathfrak{N}_{i-1}$  and  $\mathfrak{N}_i$  but not the other,  $\text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner)$ , where  $\phi$  is the formula whose Gödel number is  $n$ .

Suppose that  $\mathfrak{N}_{i-1} \models \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner)$ . Then  $\mathfrak{N}_{i-1} \models \text{Pr}_{\text{PA}}(\ulcorner \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner) \urcorner)$ .

Hence if  $m = \ulcorner \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner) \urcorner$ , then  $\mathfrak{N}_{i-1} \models \text{Pr}_{\text{PA}}(\ulcorner (\theta_m \rightarrow \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner)) \urcorner)$ , so that  $\mathfrak{N}_{i-1} \models \mathfrak{H}_{\text{PA}}(\ulcorner \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner) \urcorner) = 1$ , so that  $\mathfrak{N}_i \models \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner)$ , giving a contradiction.

So it must be the case that  $\mathfrak{N}_{i-1} \models \neg \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner)$  while  $\mathfrak{N}_i \models \text{Pr}_{\text{PA}}(\ulcorner (\theta_n \rightarrow \phi) \urcorner)$ , so that  $f_i(n) = 0$  and  $f_{i+1}(n) = 1$ .

We also note two other facts. Firstly, we must have  $n \leq k$ , since  $f_i(k) \neq f_{i+1}(k)$ . Also, we must have that for all  $j \geq i$ ,  $f_j(n) = 1$ .

So,  $1 < i < j$  implies that  $f_i < f_j$ . Moreover, there exists  $n \leq k$  such that  $f_i(n) = 0$  and  $f_j(n) = 1$ . So the functions  $f_i \upharpoonright \{n : n \leq k\}$  are all different, for  $i > 1$ , so there are only finitely many of them.  $\square$



Look up Kreisel's paper.

You get a Kripke frame for provability logic. You can carry out the Henkin process entirely inside  $\mathbb{N}$ . When you do, the models are guaranteed not elementarily equivalent—or truth would be definable.

It is a fact familiar to those who have done elementary courses in Set Theory that most familiar facts about the semiring  $(\mathbb{N}, 0, 1, +, \cdot)$  can be proved from PA. It is a fact familiar to any who have done elementary courses in Logic that there are many models of PA other than the semiring of natural numbers. However that result leaves open the possibility that all those models might be *elementarily equivalent*: that the same sentences of  $\mathcal{L}$  might be true in all of them. In this course we will see that this is very far from being the case.